

Firm Concentration & Job Design:

The Case of Schedule Flexible Work Arrangements

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Abstract

We study how employer concentration affects job design, focusing on schedule flexibility as a non-wage job attribute. Using the full text of 46 million UK job vacancies from 2014–2019, we construct new measures of flexible work arrangements using supervised machine learning and large-language-model–assisted annotations. Adopting an instrumental variable strategy using a Bartik-style prediction of the change in employer concentration based on nationwide hiring trends, we show that higher concentration reduces the value of worker compensation partly through shifts toward “risky” flexibility that exposes workers to earnings volatility.

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1 Introduction

A job is more than just a wage. From flexibility over when and where one works to the provision of health insurance and tuition assistance programs, there are many dimensions along which jobs vary. Workers and employers can have strong, and potentially conflicting, preferences over the non-wage characteristics of a job. Workers commonly cite non-monetary features as key determinants of their search behaviour ([Adams-Prassl et al., 2023](#); [Carrillo-Tudela et al., 2023](#)) and have a significant willingness to pay for a wide range of job characteristics including freedom from harassment and dignity at work ([Folke and Rickne, 2022](#); [Dube et al., 2022](#)). Non-wage dimensions of jobs can have a direct effect on firm profit; certain features may be costly to supply or have an impact on workplace productivity ([Gibbs et al., 2021](#); [Etheridge et al., 2020](#)).

Schedule flexibility is an increasingly salient dimension of job design. However, the extent to which the job flexibility offered in the labor market is a positive amenity that satisfies workers’ demand for work-life balance, as opposed to a cost-saving measure for employers looking to shift risk onto employees, is a matter of fierce policy debate. As described in an independent review of working practises for the UK government, “being able to work when you want is a good thing; not knowing whether you have work from one day to the next when you have bills to pay is not” ([Taylor et al., 2017](#)). An experimental literature highlights that the amenity value of flexibility for workers depends crucially on whether workers or employers control the work schedule ([Mas and Pallais, 2017](#); [Datta, 2019](#)). The “type” of flexibility that is realized in particular segments of the labor market is likely to depend on the relative bargaining power of firms and workers, although this has not been

examined from a theoretical nor empirical perspective.

In this paper, we assess the amenity value of schedule flexibility as it is offered to workers in practice. We construct a new measure of flexible work arrangements using the full text of more than 46 million online job vacancies posted in the UK between 2014 and 2019 and collected by Lightcast (formerly Burning Glass Technologies). Rather than relying on keyword searches for “flexibility” as this can generate false positives (e.g. references to worker traits) and false negatives (e.g. implicit descriptions of variable schedules), we adopt a supervised machine learning approach grounded in manual and large language model (LLM)–assisted annotations.

We manually annotate the work arrangements described in approximately 7,000 vacancies, which define the conceptual ground truth for our classification. As schedule flexibility is often conveyed through localized statements, we supplement these vacancy-level labels with sentence-level LLM annotations that allow us to identify flexibility-related content with high precision. Using these annotated data, we train machine learning classifiers to infer work arrangements from vacancy text at scale. The resulting measures exhibit high predictive accuracy and generate an occupational distribution of schedule flexibility that is strongly correlated with a high-quality cross-section measure of job flexibility from a nationally representative survey.¹

When piloting our approach, we found that it wasn’t possible consistently to identify whether workers or employers had control over the variation in schedules from the vacancy text of flexible jobs. For example, does a work arrangement described as a “Casual contract!

¹The survey measure was designed by [Adams-Prassl et al. \(2020\)](#) and included in the Understanding Society household panel during summer 2020.

Allows for flexibility” describe a worker or employer determined schedule?² Thus, in addition to flexibility, we also identify whether a job vacancy is salaried or not (i.e. a fixed income versus a wage contract). Conceptually, unsalaried schedule flexible jobs open the worker up to the possibility of an unpredictable, precarious income stream, because flexibility in the timing and number of hours worked can translate directly into earnings variation. These “risky” flexible jobs are in contrast to salaried flexible jobs which come with a fixed income, and where schedule flexibility only impacts the timing of work but not of remuneration. As a result, while remuneration type and schedule control are separate concepts, their impact on worker welfare is intertwined since unsalaried workers are more exposed to the consequences of employer-control over schedule. In the paper, we refer to unsalaried and salaried flexible positions as “risky flexible” and “safe flexible” jobs respectively.

We first provide new stylized facts on the characteristics of flexible jobs. We find that schedule flexibility was relatively widespread even in the years before the Covid-19 pandemic: 15% of all vacancies in the UK in 2019 were advertised as schedule flexible. Just over half of these vacancies specified “risky” flexibility. We document considerable differences between risky and safe flexible jobs. Safe flexibility is concentrated in higher-paying occupations with greater skill requirements, whereas risky flexibility is prevalent in lower-paid, lower-skill jobs. In elementary occupations, more than 20% of vacancies offer a flexible schedule, and over 80% of these positions are risky. These contrasts persist even within occupations and local labor markets: risky flexible jobs are associated with lower wages and narrower skill requirements compared to both safe flexible and fixed-schedule positions when controlling for occupation,

²Approximately 70% of manually tagged vacancies that described a schedule flexible arrangement did not clearly specify “employer” nor “employee” controlled flexibility. Furthermore, who controls the job schedule might vary over time within an employment relationship given changes in the relative bargaining power of workers and firms.

county and time period.

At the firm level, the expansion of schedule flexibility reflects both extensive and intensive margin changes. Between 2014 and 2019, the share of firms posting at least one flexible vacancy rose substantially, with growth in both in the proportion of firms posting safe and risky flexible arrangements. Flexible vacancies are disproportionately posted by firms with higher hiring volumes and by lower-wage employers, patterns driven primarily by the use of risky flexible contracts. Linking vacancy data to mandatory gender pay gap disclosures further shows that firms with smaller average gender pay gaps offer a higher share of flexible jobs.

To help motivate our analysis of the relationship between employer concentration and schedule flexibility, we develop a simple model of the equilibrium provision of non-monetary job features and wages under employer market power. We extend a simple Cournot model of oligopsonistic competition to firms making wage and non-wage (job “feature”) offers. The optimal bundle will depend on three key parameters: whether a job feature is costly or profitable for the employer to provide; whether it is an amenity or a disamenity to the worker; and the degree of market power of the employer, itself a function of employer concentration in the local labor market. The model yields predictions about the relationship between employer concentration and job features that differ according to whether a job feature is utility-enhancing or utility-diminishing for workers, and whether it is profitable or costly for firms to supply.

To test the predictions of our theoretical model and assess the amenity value of flexibility, we estimate the relationship between employer concentration and the prevalence of flexibility in a local labor market. We construct the Herfindahl-Hirschman Index (HHI) of employer

concentration using job vacancy data at the 3-digit occupation-county level as our measure of monopsony power (Azar et al., 2020). In addition to including a rich set of fixed effects to control for any systematic variation across local labor markets that might be driving the concentration-flexibility relationship, we construct a granular instrument for employer concentration (Gabaix and Koijen, 2024; Schubert et al., 2021), based on nationwide shocks to changes in firm-level hiring. The motivating idea behind this instrumental variable approach is that a firm’s local hiring will partly reflect its conditions nationally, which are unlikely to be driven by changes in local conditions.

We find a robust positive relationship between employer concentration and advertised schedule flexibility. As workers’ outside options deteriorate, firms are more likely to offer risky flexible jobs, while the prevalence of safe flexible jobs changes little with employer concentration. Interpreted through the lens of our model, these results suggest that risky flexibility is a disamenity that increases firms’ profits, while safe flexibility is an amenity. We find that the positive relationship between employer concentration and risky flexibility is driven by those occupations in which employer-controlled flexibility is most prevalent.³ This is what we would expect: in occupations where workers can control their schedules, risky flexibility is not such a disamenity. It is important to note that while advertised job characteristics that the Lightcast data capture are likely closely linked to the eventual equilibrium arrangements that emerge after bargaining and hiring, if posted schedules differ systematically from realized schedules, this could alter the link between market structure and workers’ experienced flexibility.

Our results highlight that schedule flexibility is not the same job feature for all workers.

³As measured in a one-off nationally representative survey, Understanding Society.

Risky flexibility is much more prevalent at low wages, in less-skilled occupations, and in less competitive labor markets. This feature is a disamenity for workers, although one that is productive for firms. Yet, safe flexibility is an amenity and is the dominant form of flexibility at high wages and in high skilled occupations. Our results suggest that the distribution of welfare-relevant job features accentuates overall inequality in the labor market. Furthermore, our findings suggest different mechanisms lie behind the wage penalty for safe and risky flexible jobs compared to fixed-schedule jobs. Safe flexible vacancies pay less because workers are willing to give up part of their wages for an important amenity, while risky flexible vacancies are paid less because they are more likely to be offered in labor markets with poor outside options for workers ([Hamermesh, 1999](#); [Bonhomme and Jolivet, 2009](#); [Hwang et al., 1998](#)).

This paper makes three main contributions to the literature. First, we contribute to the growing body of work on alternative and flexible work arrangements ([Mas and Pallais, 2020](#); [Datta et al., 2019](#); [Boeri et al., 2020](#); [He et al., 2021](#); [Wiswall and Zafar, 2018](#); [Maestas et al., 2023](#)). A recent strand of this literature estimates worker preferences for flexible arrangements using experimental variation and typically finds that the average worker is not willing to pay much for schedule flexibility ([Mas and Pallais, 2017](#); [Datta, 2019](#); [Schouwer and Kesternich, 2022](#)).

A key insight of our analysis is that treating schedule flexibility as a single job characteristic obscures economically meaningful heterogeneity. Much of the existing literature either studies specific forms of flexibility in controlled environments or pools together flexible arrangements that differ substantially in their exposure to earnings risk. For example, [Mas and Pallais \(2017\)](#) elicit willingness to pay for employee- and employer-controlled schedule

flexibility in a hypothetical call-center job where all positions are paid hourly. In this setting, flexible schedules mechanically generate earnings variation as hours fluctuate, so estimated willingness to pay reflects not only preferences over schedule control but also aversion to income variation. This distinction is particularly relevant for employer-discretion flexibility, where workers may face both limited control over hours and heightened earnings uncertainty.

By distinguishing between safe flexibility, which insures workers against earnings variation, and risky flexibility, which does not, we show that schedule flexibility can function either as an amenity or a disamenity depending on its contractual context. We further demonstrate that labor market concentration systematically shifts the composition of flexible work toward its riskier form, amplifying inequality in worker welfare across wages and occupations. To our knowledge, no prior empirical work has documented this distinction or linked it to employer market power.

Second, we make a theoretical and empirical contribution to the growing literature on monopsony in the labor market. While several models use firms' provision of amenities as a microfoundation for monopsony power (Card et al., 2018; Lamadon et al., 2022), our approach runs in the opposite direction: we take monopsony power as given and characterize the optimal provision of wages and non-monetary compensation under these conditions. Our theoretical framework and empirical analysis focus on changes in employer concentration that alter bargaining power without affecting the volume of job opportunities or firms' production technologies (Beaudry et al., 2012; Caldwell and Harmon, 2019; Prager and Schmitt, 2021).⁴

Complementing recent work on the effects of employer concentration on wages (Benm-

⁴In our empirical specifications, we directly control for labor demand and show no systematic relationship between employer concentration and skill requirements.

elech et al., 2020; Azar et al., 2020; Rinz, 2022; Schubert et al., 2022), we show that increased concentration also reduces non-monetary compensation. An analysis of wages alone therefore understates the welfare consequences of monopsony power. While personnel economics has long emphasized that firms optimally design job attributes (Lazear, 2000; Oyer and Schaefer, 2005), there is little empirical evidence on how amenities adjust in response to changes in labor market power. We show that increases in employer concentration affect the provision of job amenities, shifting flexibility toward arrangements that expose workers to greater earnings risk. The closest paper to ours is Dube et al. (2022), who study the relationship between monopsony power and dignity at work, with a focus on how amenities mediate the effects of minimum wage changes. Our papers are complementary but distinct: they emphasize different microfoundations for monopsony power and pursue different empirical questions.⁵

Finally, we demonstrate that online job vacancy text is a rich and underutilized source of information on non-wage job characteristics (Audoly et al., 2024). Prior work has used vacancy data to study skill demand (Hershbein and Kahn, 2018; Deming and Kahn, 2018), hiring responses to policy (Marinescu, 2017; Duchini et al., 2020), and economic shocks (Forsythe et al., 2020). We are among the first to use machine learning methods to exploit the full text of vacancy postings to measure flexible work arrangements at scale, rather than relying on keyword searches or job titles alone. While recent work applies frontier natural language processing techniques to related questions (Hansen et al., 2023), we show that a simpler, computationally efficient classifier achieves comparable accuracy for flexible scheduling, validating our measurement strategy.

⁵Dube et al. (2022) focus on horizontal job differentiation, while we model employer market power as a function of market concentration. Their empirical analysis centers on policy-induced wage floors, ours examines how variation in employer concentration shapes the composition of job attributes in equilibrium.

The rest of this paper proceeds as follows. In Section 2, we introduce the job vacancy data, describe our classification of schedule flexible jobs, and outline the machine learning algorithm used to identify the job features specified in the vacancy text. In Section 3, we summarize the main descriptive patterns of flexible vacancies in our data and describe the differences between safe flexible, risky flexible, and fixed-schedule vacancies. In Section 4 we develop a simple model of endogenous job feature provision under monopsony. In Section 5 we analyze the relationship between changes in employer concentration and different types of flexibility to identify the amenity value of these job features. Section 6 concludes.

2 Measuring Flexible Work Arrangements

There is limited consistent time series evidence on the prevalence of schedule flexibility, let alone whether variation in work schedules is controlled by the worker or employer (Adams et al., 2018; Abraham and Amaya, 2019; Mas and Pallais, 2020). To make progress, we use two novel sources of data on flexible work arrangements. We analyze the full text of the near-population of job vacancies advertised online in the UK labor market from Lightcast and identify the characteristics of the advertised work arrangements using supervised machine learning methods. This yields a measure of schedule flexibility that is consistent across occupations, local labor markets, and time. We complement our vacancy-based flexibility measure with (one-off) high quality, representative data on job flexibility from Understanding Society, a large UK household survey. This allows us to check the quality of our vacancy-based measure and to provide additional insights into the characteristics of schedule flexible jobs that are not captured by vacancy data.

2.1 Data

Lightcast Data. We use the UK NOVA data feed from Lightcast as our corpus of job vacancies. This is a widely used database that aims to cover the universe of online job postings. Since 2012, Lightcast has collected the near-universe of electronically posted job vacancies in the UK by scraping 7,500 online job boards and company web pages. We restrict our analysis to data from 2014; between 2012 and 2014, the number of webpages that Lightcast scraped for job adverts increased rapidly. Since this date, the sources of vacancy information has been stable. Our analysis is based on the full vacancy text of over 46 million unique online job postings between 2014 and 2019. Remuneration information is extracted from job postings using text and number patterns. If wages are specified in the vacancy text, then standardized language is used to specify the salary band. For example, the wage can be stated in a section titled “Salary”, or “Remuneration Structure”. Skills are identified with keyword patterns, which represent inclusion and exclusion criteria. For example, the keyword *python* would appear as a computer programming skill if it is followed by *programming*, but not if it is followed by *hunter*.

As with all studies that make use of job vacancy data, it is important to note that the data comes with some caveats. The source only captures jobs advertised online. However, some jobs are not advertised at all and others may only be advertised on offline or closed portals. Table A.4 gives the headline summary statistics for the Lightcast data and their closest counterparts from representative surveys. Lightcast captured 83% of all vacancies posted in the UK over this time period (as given by the Office for National Statistics’ official vacancy count). The Lightcast wage distribution closely aligns with the distribution of wages of

employees (i.e. a stock measure) in the Annual Survey of Hours and Earnings (ASHE), a 1% sample of employee jobs in the UK.⁶ 37% of Lightcast vacancies omit wage information but we demonstrate there are no statistically significant observable differences between vacancies with and without posted wages (Table A.5). Our main empirical specification will rely on variation in firm concentration and schedule flexibility within occupation-county cells over time. Figure A.7 analyzes whether there has been any change in the representativeness of Lightcast data over time within these cells. It shows that there has been no change in the relationship between the number of vacancies in each occupation-county cell (from Lightcast data) and the number of employees in the same cell as given by ASHE.⁷

Understanding Society. To check the quality of our measure of flexibility, we use one-off survey data from Understanding Society, one of the largest household panel studies in the world. The main panel is known to provide a strong basis for population inferences (Crossley et al., 2021). In response to the Covid-19 pandemic, the survey introduced a special web-based survey in April 2020. All individual members of households enrolled in the existing panel who were aged sixteen or over in April 2020 were invited to participate in the Covid-19 study.

We make use of responses of individuals to the April to September 2020 waves of the Covid-19 study who reported that they were employed in January or February 2020 (i.e.

⁶The share of jobs that are permanent in the Lightcast data is considerably less than in ASHE, but this likely reflects the fact that by their very nature, temporary jobs have a higher churn rate and so are overrepresented in the flow of vacancies than the stock of employees.

⁷Specifically, we regress the number of vacancies in each occupation-county cell on the number of employees in the same cell as given by ASHE. We run this regression separately for each year of Lightcast data, and plot the resulting R^2 , which allows us to see whether the predictiveness of ASHE for Lightcast changes over time. We also run the analysis for median wage in each occupation-county cell. Both panels of Figure A.7 show that this R^2 stays flat over time.

pre-pandemic).⁸ These waves included the following retrospective question on job flexibility and schedule control for a respondent’s main pre-pandemic job:⁹

“How were your hours set during January and February?”

- *Fixed weekly hours;*
- *I chose my hours;*
- *My employer chose my hours, with a minimum guaranteed number of hours;*
- *My employer chose my hours, with no minimum guaranteed number of hours.”*

We make use of answers to this question, in addition to survey responses concerning other job characteristics such as occupation, remuneration type (salaried versus non-salaried), and wage to validate our vacancy-based approach. We cannot use this data for our main analysis as the question was only collected over summer 2020 and there is insufficient sample size for analysis at the local labor market level.¹⁰

2.2 Identifying Schedule Flexibility in Vacancy Text

Our goal is to retrieve all vacancies that describe flexible work arrangements from the set of Lightcast job adverts. [Duchini et al. \(2020\)](#) is the only existing paper we are aware of

⁸The full sample covers 12,695 respondents. We only keep those who report they were in-work in January or February 2020 (dropping 1,710 respondents). We then drop 861 observations with missing occupation information. Finally, we drop 46 observations that had missing responses to the job flexibility question. These leaves an analysis sample of 9,688.

⁹This question was piloted in the real-time surveys collected by [Adams-Prassl et al. \(2020\)](#).

¹⁰We explored alternative survey data sources, including the 2015 European Working Conditions Survey (EWCS), which contains detailed questions on job flexibility and aligns well with the timing of our main analysis. However, the EWCS sample is substantially smaller than that of Understanding Society (1,337 employed respondents versus 9,688), and it does not permit a distinction between salaried and non-salaried workers.

that studies schedule flexibility from vacancy text; they classify a vacancy as flexible if a particular set of words are present in the advert. A limitation of this approach is that it is subject to both false positives and false negatives. For example, the word “flexible” can be used to describe a personality trait rather than a work arrangement (“We are looking for a motivated, flexible & committed individual”). A large number of words are used to describe flexible arrangements making it hard to specify the full dictionary a priori (“Be your own boss! Set your own schedule.”). Further, schedule flexibility is often conveyed through short, localized statements such as references to rotating shifts, variable hours, on-call duties, or overtime expectations rather than through repeated or standardized language across an entire posting.

To address this challenge, we adopt a sentence-level supervised classification approach grounded in manual annotation and supplemented by large language model (LLM)–assisted labeling. We define a schedule flexible work arrangement as one in which the timing of work is not fixed in the contract and is left open to future (possibly repeated) negotiation between the employer and the employee. This definition encompasses both worker- and employer-driven flexibility; when piloting our approach we could not consistently infer who controlled variation in the schedule from vacancy text (see Section 2.3).

We begin by constructing a ground-truth dataset through manual annotation of job vacancies drawn from the Lightcast corpus. Appendix D gives full details of our procedure. In summary, annotators classify postings according to whether they contain any reference to schedule flexibility, following a common written protocol. Given the impossibility of fully enumerating all the possible ways in which schedule flexibility might be described, a subset of 1,700 vacancies in the training set were annotated by two readers independently to ensure

consistency of our definition.¹¹ As scheduling information is often expressed at the sentence level, we then segment vacancy text into sentences and treat sentences as the unit of analysis for identifying flexibility-related content.

To scale the training data beyond what was feasible and cost-effective with manual annotation alone, we augment the human-labeled dataset with LLM-based sentence-level annotations generated under the same conceptual definition. This allowed us to generate an additional 49,907 labelled sentences to train our classifier, 6,703 of which contained a reference to schedule flexibility.¹²

Using this annotated data, we estimate a sentence-level text classifier implemented using *fastText*. We represent sentences as order-invariant bag-of-words features, with word-level bigrams included as a tunable option. This representation is then passed to a linear classifier that predicts the probability of the positive class using a binary logistic loss. Model hyperparameters — including the learning rate, number of training epochs, embedding dimension, and the inclusion of word-level bigrams — are selected via grid search with cross-validated balanced F-measure at the sentence level. We aggregate sentence-level predictions to the vacancy level using an any-sentence criterion: a vacancy is classified as schedule flexible if any of its constituent sentences is predicted to indicate schedule flexibility. This aggregation rule reflects the fact that a single statement is sufficient for flexibility to be a meaningful attribute of the job. Full details are provided in Appendix Appendix D.

¹¹At times, the vacancy text can contain ambiguous or contradictory statements about whether the vacancy is of a particular work arrangement. Having two researchers label each vacancy independently and then reconcile any disagreements in a mutual discussion, potentially involving the senior author on this paper, also provided a mechanism for addressing this ambiguity.

¹²Applying LLMs directly to the full set of vacancy postings would have been prohibitively expensive relative our research resources at prevailing computational and usage costs, motivating their use as a targeted complement rather than a substitute for human classification.

2.3 Identifying Other Employment Characteristics

Control over schedule. When piloting our approach to develop instructions for annotators, it was not possible for us to consistently identify whether workers or employers had control over the variation in schedules from the vacancy text. Approximately 70% of manually tagged vacancies that described a schedule flexible arrangement did not clearly specify “employer” nor “employee” controlled flexibility. For example, does a work arrangement described as a “Casual contract! Allows for flexibility” describe a worker or employer determined schedule? Even promises of “work as much or as little as you need” may be subject to managerial approval, and demands of “working according to the needs of the business” will eventually need to be agreed to by the worker.

Responses to the Understanding Society survey also provide grounds for doubting a clean distinction between employer versus employee control in vacancy text exists. Appendix Figure A.1 (a) gives the prevalence of different forms of schedule flexibility in the survey data. 24% of employed respondents reported having variable hours. Of those, 47% report having control over their schedule, and a further 36% report that their employer has control over their schedule but *with* guaranteed hours. Only around one in six of those employed on flexible contracts report having employer control over the schedule *without* minimum guaranteed hours. Thus, in practice, which party has control over the schedule is likely to lie on a continuum, rather than being represented by a simple binary.

Remuneration Type & Permanency. While variation in hours as such can be an important component of workers’ utility, concerns about employer-controlled flexible work arrangements typically relate more to the earnings volatility such arrangements can generate. This

motivates the distinction between different flexibility types explored in this paper. Schedule flexibility translates into earnings variation primarily for workers on unsalaried contracts, where pay is directly tied to hours or output rather than fixed over time. Data from Understanding Society show a strong association between remuneration type and schedule control. Workers who can choose their own hours are substantially more likely to be salaried, while those without guaranteed hours are rarely paid a salary (Appendix Figure A.1 b). Thus, although remuneration type and schedule control are not perfectly aligned, they are closely related in practice.

Motivated by this relationship, we also classify vacancies by remuneration type, using earnings information reported in 63% of postings. Positions advertising annual pay are classified as salaried, while those listing hourly or daily pay are classified as unsalaried. We additionally annotate whether vacancies describe permanent contracts as a further positive job amenity.

Machine Learning Approach Whether a job is permanent or salaried is typically described more consistently across vacancy postings than schedule flexibility information. These attributes are therefore well suited to vacancy-level as opposed to sentence-level classification methods as the more standardized vocabulary and phrasing patterns make these attributes easier to identify through document-level keyword presence rather than requiring sentence-level semantic analysis.

For these characteristics, we represent each vacancy using a bag-of-words representation constructed from the full posting text. We include uni- and bigrams and selected higher-order n-grams, remove common stop words, and restrict the vocabulary to a fixed size

to limit overfitting. With our vocabulary in hand, we represent vacancies via their text descriptions with a binary term–document incidence matrix. We apply a logistic regression model with Least Absolute Shrinkage and Selection Operator (LASSO) regularization to the incidence matrix. The regularization parameter for each model is chosen via cross-validation, again maximizing the balanced F-measure. Once trained, the models are applied to the full Lightcast corpus to generate vacancy-level predictions for permanent contracts and salaried pay. Full details are provided in Appendix Appendix D.

2.4 Model Performance and Validation

Across all employment characteristics, the trained models achieve high precision and recall, substantially outperforming keyword-based benchmarks evaluated on the same test data. Table 1 shows the (out-of-sample) precision, recall and balanced F -measure for each job feature.¹³ Our model demonstrates consistently strong performance across all the dimensions of work arrangements that we predict, both in absolute terms and when compared to a benchmark approach using the five most positively weighted keywords identified by machine learning models. This benchmark is quite generous to the keyword search approach, as these keywords have been chosen to maximize the performance of the text classification and it is uncertain whether they would have been chosen in a manual keyword selection.

¹³These terms are defined as follows:

$$\begin{aligned} \text{F-score} &= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \\ \text{Precision} &= \frac{\sum_i (\hat{c}_i = 1 \ \& \ c_i = 1)}{\sum_i (\hat{c}_i = 1 \ \& \ c_i = 1) + \sum_i (\hat{c}_i = 1 \ \& \ c_i = 0)} \\ \text{Recall} &= \frac{\sum_i (\hat{c}_i = 1 \ \& \ c_i = 1)}{\sum_i (\hat{c}_i = 1 \ \& \ c_i = 1) + \sum_i (\hat{c}_i = 0 \ \& \ c_i = 1)} \end{aligned}$$

with $\hat{c}_i$ representing the predicted classification of vacancy i and c_i the actual classification.

Table 1: Prediction Accuracy of Trained Logistic Regression Model

Contract type	Logistic Regression Model			Δ Keyword
	Precision	Recall	F	F
Schedule flexible	0.9118	0.9215	0.9167	0.2719
Permanent	0.9294	0.9736	0.9510	0.0223
Salaried	0.8604	0.8415	0.8503	0.2070

Notes: Improvement to keywords, Δ Keyword, is defined as the difference in the balanced F-score when our ML approach is used over a keyword search. The schedule flexible category shows sentence-level classification performance.

Validation We validate our measures of job features by comparing them to the survey-based measures from the Understanding Society survey data. Specifically, we compare the prevalence of flexible work arrangements in three-digit occupation codes in Lightcast in 2019 to their prevalence in the Understanding Society survey in January 2020. Appendix Figure A.2 (a) shows that there is a strong positive relationship between these measures. A regression of the Lightcast flexible vacancy share on its corresponding survey measure yields a significant positive point estimate of 0.4 (p-value < 0.0001). Appendix Figure A.2 (b) shows that there is also a strong and significant relationship between the prevalence of salaried positions across the two datasets.

3 Descriptive Facts

Before analyzing the relationship between schedule flexibility and employer concentration, we provide new stylized facts on the prevalence and characteristics of flexible work arrangements. We restrict the sample to the set of job ads that have wage information. This allows us to keep the sample stable as we analyse the interaction of schedule flexibility with remuneration type. In Table A.5, we show that there is no statistically significant difference in the non-wage characteristics between vacancies that do and do not post a wage. We refer to vacancies that are schedule flexible and unsalaried as “risky flexible” jobs and those that are schedule flexible and salaried as “safe flexible” jobs. In 2019, 14.9% of all vacancies in the UK are flexible, with an equal split between safe and risky flexibility.

3.1 Vacancy-Level Characteristics

Figure 1 gives the overall share of flexible vacancies over time, across occupations, the number of skills a vacancy requires, and the wage distribution. The proportion of schedule flexible vacancies has grown significantly over time from 7% in 2014 Q1 to 14.9% in 2019 Q4 (panel (a)). The growth in risky flexibility has been slightly greater than safe flexibility. While risky flexibility accounted for 47.6% of all schedule flexible vacancies in 2014, this rose to 51.4% in 2019.¹⁴

Figure 1 (b) demonstrates that the prevalence and type of flexibility varies significantly across occupations. The share of flexible vacancies varies from 6.3% amongst managers to

¹⁴Appendix Figure A.3 shows the over time change in the proportion of flexible jobs for vacancies posting a wage above and below average. The prevalence of schedule flexibility grew for both low and high wage positions, with growth in risky flexibility faster in low wage jobs and growth in safe flexibility greater for high wage jobs.

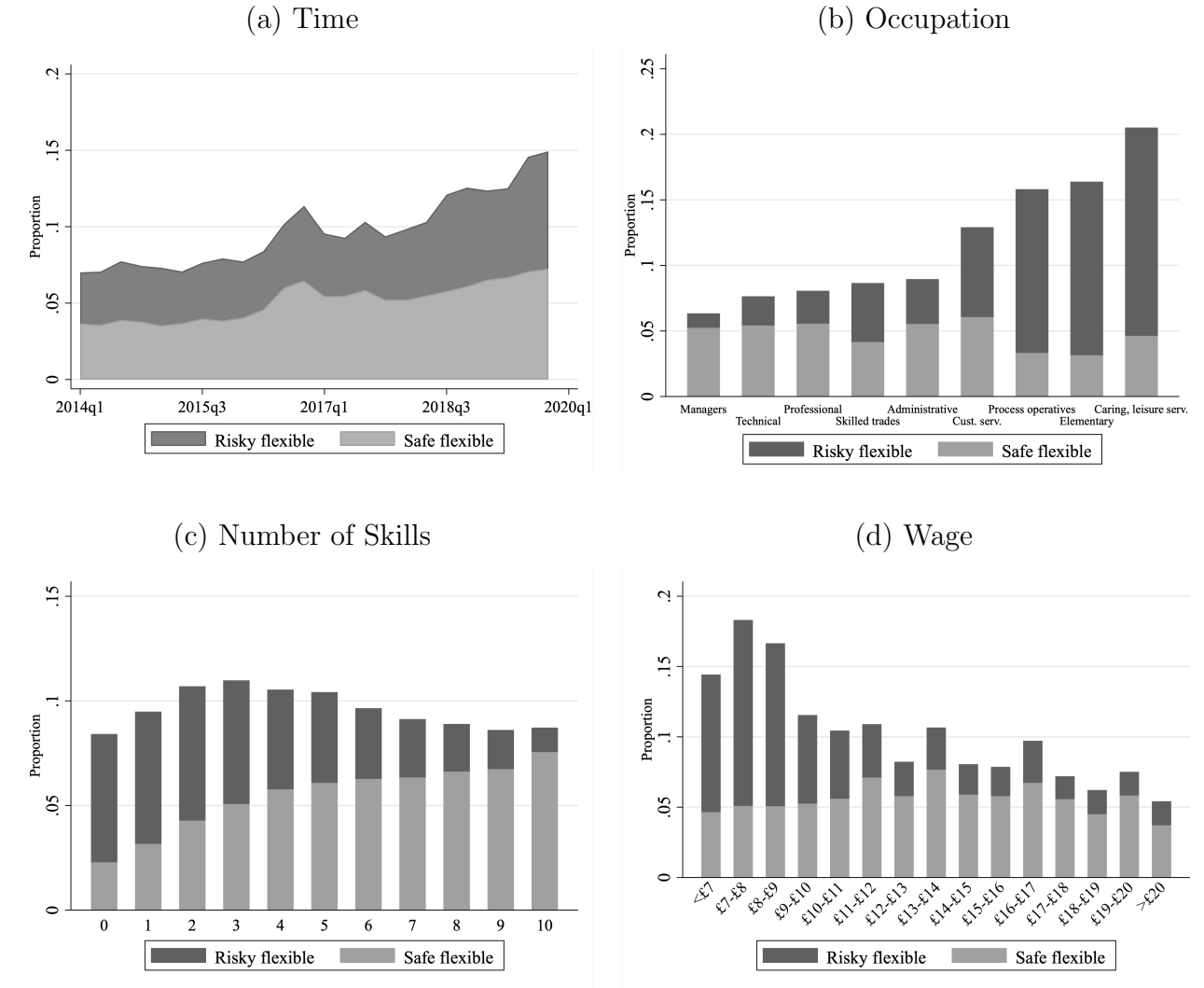
20.5% in caring and leisure services. Flexibility in lower-skill occupations is much more likely to be risky rather than safe: in elementary occupations (which include cleaning, low-skilled farm work, and security guards), only about 19.9% of flexible jobs come with a salary.

Figures 1 (c) and (d) make use of measures that Lightcast themselves have extracted from the vacancy text. Figure 1 (c) shows the prevalence of flexible vacancies by Lightcast’s measure of the number of skills required by an advert. The prevalence of risky and safe flexibility varies significantly across the skill distribution. Risky flexibility is much more common amongst adverts that specify only a small number of skills. The opposite is true for safe flexibility. Lightcast also extracts and standardizes information on hourly wages from job adverts that include remuneration information. Risky flexibility is much more prevalent amongst low wage jobs (Figure 1 (d)). Approximately 16.9% of vacancies paid less than £9 per hour are flexible, and the majority of these positions are risky flexible jobs. For jobs paying more than £20 per hour, only 5.4% of the positions are described as flexible. This gradient is driven by changes in the prevalence of risky flexibility across the wage distribution.

To assess whether the patterns in Figure 1 reflect differences in the occupational composition of flexible and non-flexible jobs, or instead differences in the prevalence of salaried versus non-salaried contracts, Table 2 reports results from descriptive regression models in which vacancy characteristics are regressed on indicators for job flexibility, non-salaried pay, and their interaction. All specifications include three-digit occupation, county, and time fixed effects.

Focusing first on wages, we find that—conditional on occupation, location, and time—flexible jobs are associated with a statistically significant wage penalty, with the lowest wages observed in risky flexible jobs (flexible and non-salaried). Table 2 also reveals strong relation-

Figure 1: Flexible Vacancies and Other Job Characteristics



Notes: Lightcast data from 2014-2019. The sample is restricted to job adverts with an occupation code, wage, and county. Wages deflated to the 2019 price level.

ships between flexibility and skill requirements. Relative to fixed-schedule salaried jobs, safe flexible jobs are associated with a broader set of required skills, whereas risky flexible jobs are associated with a narrower set.

Columns (4) and (5) examine the interaction between flexibility and job permanence. Among salaried jobs, flexible positions are more likely to be temporary than their non-flexible counterparts, while among non-salaried jobs the reverse is true. This pattern raises questions about the substantive meaning of “permanent” status in risky flexible jobs, given that employers can effectively provide zero hours indefinitely without formal dismissal. Consistent with this interpretation, policy and legal commentary has questioned whether the distinction between “permanent” and “temporary” contracts is meaningful in the context of low-wage flexible work ([Adams et al., 2018](#)).

3.2 Firm-Level Characteristics

We now turn to firm-level patterns in Figure 2. For this analysis, we aggregate outcomes to the firm–county level and restrict the sample to vacancies with a recorded firm name in the Lightcast data. Figure 2 (a) reports the proportion of firms posting at least one vacancy with a given flexibility feature in 2014 and 2019. Over this period, the share of firms offering at least one flexible vacancy rose from 17.6% to 26.3%, with growth in both “safe” and “risky” flexible employers.

Figure 2 (b) and (c) show how the share of flexible vacancies varies across firms by vacancy volume and average wage levels, respectively. We focus on firms posting at least one vacancy in 2017, the midpoint of the sample period. Flexible vacancies are more likely to

Table 2: Flexibility and Vacancy Characteristics

	Log Wage	N. of skills		Temporary	
	(1)	(2)	(3)	(4)	(5)
Flexible=1	-0.0618*** (0.00143)	0.699*** (0.0216)	0.722*** (0.0215)	0.0324*** (0.00143)	0.0334*** (0.00140)
Non-Salaried=1	-0.0476*** (0.00406)	-1.114*** (0.0160)	-1.096*** (0.0155)	0.454*** (0.00418)	0.455*** (0.00423)
Flexible=1 × Non-Salaried=1	0.0337*** (0.00381)	-0.537*** (0.0241)	-0.549*** (0.0241)	-0.0941*** (0.00296)	-0.0946*** (0.00295)
Log Wage			0.373*** (0.0191)		0.0156*** (0.00245)
Constant	2.585*** (0.00140)	5.049*** (0.00934)	4.085*** (0.0475)	0.142*** (0.00121)	0.101*** (0.00678)
Observations	16134476	16134476	16134476	16134476	16134476
R^2	0.3553	0.2080	0.2089	0.2716	0.2717

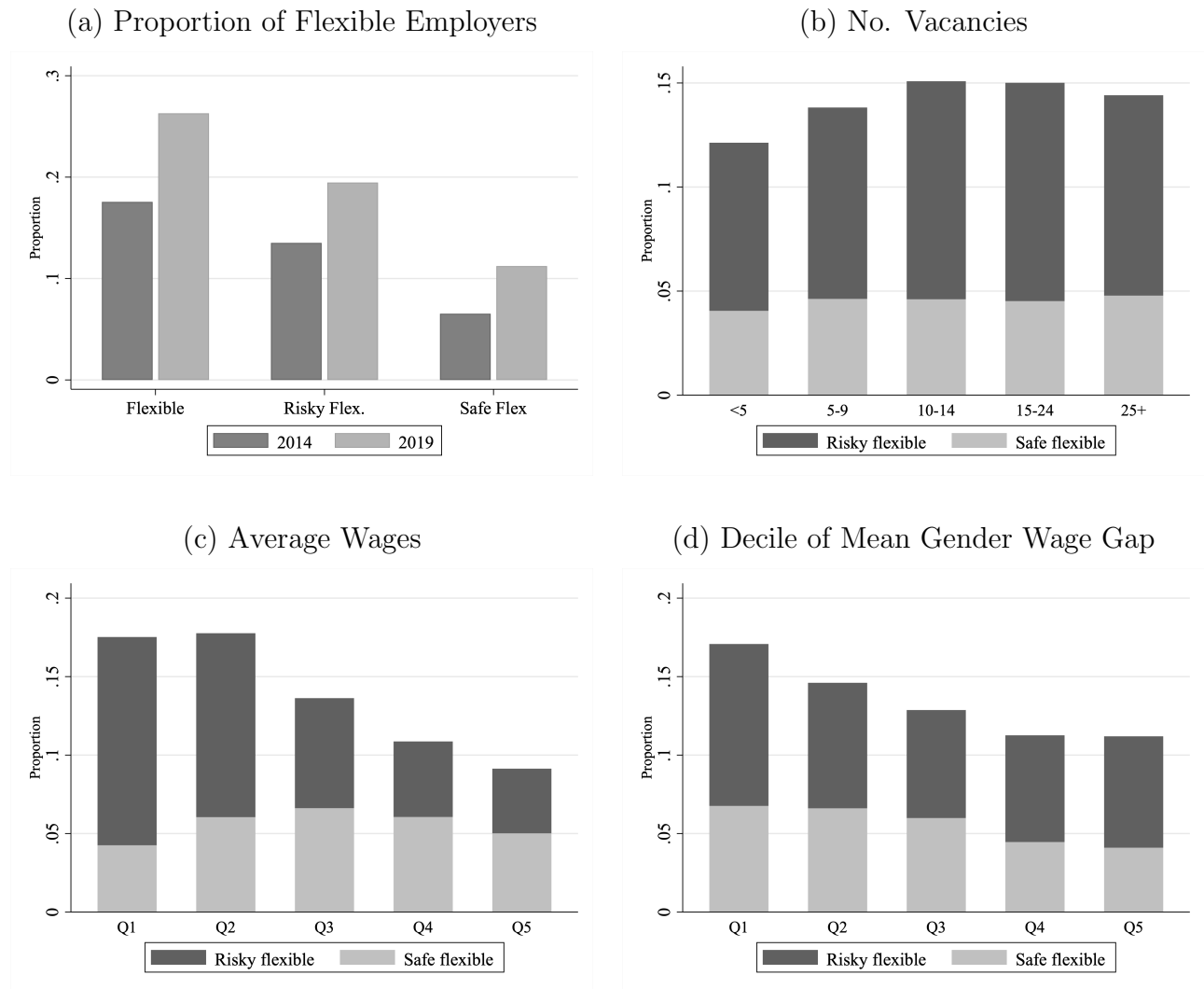
Notes: Standard errors clustered at county-occupation level in parentheses. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$
Sample restricted to vacancies with information on wage, county, and occupation. Wages are deflated and in 2019 prices. All specifications control for county, year-quarter, and occupation fixed effects.

be posted by firms with higher vacancy volumes, a pattern driven primarily by an increase in risky flexible positions. Similarly, firms in the lowest two quintiles of average wages post a substantially higher share of flexible vacancies, again largely reflecting greater use of risky flexible contracts.

Finally, Figure 2(d) links the Lightcast vacancy data to external information on firms' gender pay gaps. Since 2017, UK firms with more than 250 employees have been required to report gender pay gap statistics.¹⁵ In that year, just over 10,500 firms reported an overall measure of the average wage difference between female and male workers. Because these statistics are reported at the firm rather than establishment level, we aggregate vacancies to the firm level in 2017 and merge in the mean gender pay gap by matching firm names in Lightcast to the publicly available reporting data. We successfully match 81% of reporting firms to vacancy data in 2017. Panel (d) shows that firms with the lowest mean gender pay gaps offer a higher proportion of flexible vacancies. Notably, these firms are equally likely to offer risky and safe flexibility, suggesting that at least some of the advances in reducing the gender pay gap may come at the cost of greater precarity of workers' incomes.

¹⁵See [Duchini et al. \(2020\)](#) for an analysis of the impact of the policy on prevailing wage rates.

Figure 2: Flexible Vacancies and Other Job Characteristics



Notes: Lightcast data from 2014-2019. The sample is restricted to job adverts with an occupation code, wage, and county. Wages deflated to the 2019 price level. Panel (d) is restricted to firms with more than 250 workers whose name can be matched to UK's compulsory gender pay gap statistics.

4 Theoretical Framework

To help with the interpretation of the empirical relationship between employer concentration and schedule flexibility, we develop a model of the equilibrium provision of non-monetary job features and wages under employer market power. We extend a simple Cournot model of oligopsonistic competition to homogeneous firms making job offers in terms of a specific $\{w, f\}$ bundle. The optimal bundle will depend on three key parameters: whether job feature f is costly or profitable to the employer; whether it is an amenity or a disamenity to the worker; and the degree of market power of the employer, itself a function of employer concentration in the local labor market. The model yields predictions about the relationship between employer concentration and job features that differ according to whether a job feature is utility-enhancing or utility-diminishing for workers, and whether it is profitable or costly for firms to supply.

Model Setup Jobs are characterized by their wage, w , and a continuous non-monetary job feature, f . Workers and firms have preferences defined over bundles of job characteristics, $\{w, f\}$. For simplicity, we assume that firms and workers are homogeneous. Workers are characterised by the utility function $u(w, f)$. While utility is always strictly increasing in the wage ($u'_w > 0$), a job feature can be either an amenity ($u'_f > 0$) or a disamenity ($u'_f < 0$). A firm's profit function is given by:

$$\pi = y - w + \delta(f) \tag{4.1}$$

Figure 3: Worker and Employer Preferences over Job Features

		Employer Preferences	
		+ Profit	- Profit
Worker Preferences	+ Utility	Profitable Amenity	Costly Amenity
	- Utility	Profitable Disamenity	Costly Disamenity

where y gives the output of the match and $\delta(f)$ captures the relationship between the job feature and firm profits. The provision of a job feature can be profitable or costly. If it is profitable, we assume $\delta(f)$ is a positive concave function: $\delta(f) \geq 0$, $\delta'(f) > 0$, $\delta''(f) < 0$ on the interval $[0, b]$ where b is a positive value of f .¹⁶ If it is costly, $\delta(f)$ is a negative convex: $\delta(f) \leq 0$, $\delta'(f) < 0$, $\delta''(f) < 0$ for $f \in [0, -\infty)$.

Figure 3 illustrates the four types of job features that emerge from the different combinations of worker and employer preferences: costly amenities, profitable amenities, profitable disamenities, and costly disamenities. In practice, absent legal obligations, costly disamenities should not exist in the market place because neither the firm nor the worker benefits from their provision.

We model the labor market as oligopsonistic. It is characterized by an upward-sloping labor supply curve and a finite number of firms N competing on quantity (Cournot compe-

¹⁶ $b < \infty$ is necessary for an interior solution of profitable job features such that $\delta'(f) < 0$ when $f > b$. We discuss this in greater detail below.

tition). We assume that firms are homogenous in size and so each firm employs $1/N$ of the market. Upward-sloping labor supply could be for example motivated by frictions to geographic or occupation mobility, giving rise to local labor markets. However, for the purpose of this model, we take the slope of the labor supply curve as well as the distribution of firm sizes as given.¹⁷

Optimisation problem The firm decides how many workers to employ given the slope of the labor supply curve. Because the firm is large, its labor demand has an effect on market equilibrium and the firm internalises this effect when making its optimal decision. In a standard Cournot model of oligopsony, the firm chooses the level of its labor demand l as a function of equilibrium wage w , which in turn depends on overall market labor demand L .

This representation is not useful here because labor supply is not uniquely determined by wages: worker utility depends jointly on wages w and job features f , so multiple $\{w, f\}$ pairs can deliver the same utility and attract the same number of workers. We therefore retain the Cournot logic but pose the firm’s choice in $\{w, f\}$ space: firms choose employment indirectly through wages and job features, taking into account the fact that their hiring affects the equilibrium utility $U(w, f)$, the analogue to market wage w in the standard model.¹⁸

An oligopsonist’s profit-maximisation problem is:

$$\max_{w, f} l(U(w, f))[y - w + \delta(f)]. \quad (4.2)$$

¹⁷This is different from models in which firms’ market power stems from the specific amenities they offer (and in which these amenities are taken as exogenous). In reality, there is likely a simultaneous relationship between a firm’s market power and the amenities it provides. This feedback loop is outside the scope of this paper.

¹⁸Formally, this is a Cournot game in price space – firms take rivals’ offers as given while internalizing the aggregate feedback transmitted by market clearing (*Bresnahan 1981; Klempere and Meyer 1989*). By contrast, a wage-posting game with fixed rivals’ wages would correspond to Bertrand competition.

Solution The first-order conditions are:

$$\text{FOC}_w : \quad w = [y + \delta(f)] \frac{\epsilon \eta_w}{\epsilon \eta_w + \lambda} \quad (4.3)$$

$$\text{FOC}_f : \quad y - w = -\frac{\lambda}{\epsilon \eta_f} f \delta'(f) - \delta(f) \quad (4.4)$$

where $\epsilon = \frac{UL'(U)}{L}$ is the elasticity of labor supply with respect to market utility, $\eta_w = \frac{wU'(w)}{U}$ is the elasticity of utility with respect to wages, and $\eta_f = \frac{fU'(f)}{U}$ is the elasticity of utility with respect to amenities. Under firm homogeneity, the optimal job bundle also depends on $\lambda = \sum(1/N)^2 = 1/N$, the measure of market concentration defined as Herfindahl-Hirschman Index. Larger λ therefore corresponds to stronger employer power, while as $\lambda \rightarrow 0$ the model collapses to the competitive benchmark.¹⁹

Combining these two FOCs delivers the familiar optimality condition which pins down the wage-amenity trade-off as a function of marginal cost (or profit) of f and workers' relative preferences:

$$\frac{w}{f} = -\delta'(f) \frac{\eta_w}{\eta_f}. \quad (4.5)$$

An interior solution for a profitable amenity requires $\delta'(f) < 0$ over some range. Under the assumption that $\delta(f)$ is positive and concave on the interval $[0, b]$, this implies the existence of a “bliss point” for profitable amenities at $f = b$ beyond which the marginal profit turns negative. This also implies that, holding other parameters fixed, the optimal f is higher when $\delta(f)$ is profit-enhancing than when amenities are purely costly.

The key result of this model is the equilibrium relationship between the amount of job

¹⁹Note that λ enters the first order conditions as each firm's individual market share. Under homogeneity this coincides with the HHI. With heterogeneous firms, λ and HHI do not necessarily coincide.

feature f and labor market concentration λ . We summarize this result in the following Proposition.

Proposition 1. The optimal quantity of a job feature f decreases in labor market concentration λ when the job feature is an amenity and increases when the job feature is a disamenity. This relationship is stronger for costly amenities than for profitable amenities. *See Appendix B for the full proof.*

Proposition 1 suggests that a job feature’s “type” can be identified from its relationship with labor market concentration via its impact on employer market power. Intuitively, higher market concentration increases employer’s oligopsonistic power, which leads to lower utility for the worker. The employer achieves this by offering fewer utility-increasing amenities and more utility-lowering disamenities. For amenities, the size of this reduction depends on whether the amenity is costly or profitable. Costly amenities are similar to wage remuneration in that offering less of either directly increases firms’ profits. However, the situation is different when the amenities are profitable as the employer benefits too as long as $f \leq b$. As a result, the firm will prefer to offer f near this bliss point, leading to a relatively smaller adjustment in f compared to costly amenities.

The result of Proposition 1 carries through when the employer offers more than one job feature at a time, under some conditions. Consider a situation where labor market concentration decreases so that the firm’s oligopsonistic power falls. This implies higher market utility for workers. The employer will want to deliver this higher utility at the lowest possible cost, so it will increase the provision of the amenities with the smaller marginal cost,

or decrease the provision of disamenities with the smaller marginal profit. The absolute magnitude of the adjustment in a given job feature will thus increase in the size of the marginal change in δ : the employer would adjust the cheapest amenity the most, and the most expensive amenity only marginally, if at all. As a result, the results of Proposition 1 will hold, albeit with an attenuation effect, as long as there are no negative cross-derivatives in the marginal profitability (and utility) of the offered job features.²⁰ An example of such a setting is when utility and δ is independent of other job features.

5 Schedule Flexibility and Employer Concentration

5.1 Defining Employer Concentration

We use a vacancy-based Herfindahl-Hirschman Index (HHI) to measure employer concentration (Azar et al., 2020; Benmelech et al., 2020; Rinz, 2022). Specifically, we define a local labor market as the intersection of a county (of which there are 206) and occupation (we use the SOC 3 digit classification, comprised of 89 occupations). Employer concentration is then measured as:

$$HHI_{o,c,t} = \sum_i^I \left(\frac{v_{i,o,c,t}}{\sum_j v_{j,o,c,t}} \right)^2 \quad (5.1)$$

$$= \sum_i^I s_{i,c,o,t}^2 \quad (5.2)$$

²⁰Proposition 1 would not hold if the presence of one job feature changes workers' preferences over another job feature, for example if the provision of work from home makes workers dislike short commutes. Similarly, Proposition 1 would fail if the addition of one job feature changed the profitability of another job feature, such as if the provision of work from home made offering office space profitable.

where $HHI_{o,c,t}$ is the Herfindahl-Hirschman Index in county c and occupation o at time t , $v_{i,o,c,t}$ gives the number of vacancies that firm i posts in that county-occupation-time cell and $s_{i,o,c,t}$ is firm i 's percentage share of total vacancies in that cell. Time is measured at a quarterly frequency.²¹ This measure directly corresponds to the structural construct in our theoretical model, where the size and number of employers partly determines the degree of monopsony power employers exert over workers.

Appendix Table A.1 gives descriptive statistics on firm hiring and employer concentration in our sample. On average, 691 vacancies were posted in each local labor market (county x occupation) (29.4 per quarter). The average HHI is 2740, which is equivalent to about 3 equal-size employers per local labor market. The US Department of Justice defines a highly concentrated market as one with HHI above 2,500. In the UK, this definition applies to 37% of market-time units, somewhat below the 60% in Azar et al. (2020). There is a single employer in 24% of market-time observations.

The HHI index is not without problems. Not all jobs are advertised online and workers may search beyond the 3-digit SOC-county boundary. In principle, our measure could overestimate concentration if there are many employers who do not post vacancies online or underestimate concentration if there are significant differences in the hiring to job posting rate among small and large employers (Azar et al., 2020; Schubert et al., 2022). As discussed in Section 2, Lightcast covers 83% of UK vacancies and there has been no change in the representativeness of the source across occupation-county cells over the period considered. As a direct robustness check, we compare our Lightcast-based measure with HHI estimated from

²¹For this analysis, we use vacancies with information on firm name, county, and occupations. We drop vacancies posted by the National Health Service, and treat Greater London as a single location since not all London vacancies come with an exact borough location.

the Annual Survey of Hours and Earnings, an employer survey of a random 1 per cent of employee jobs.²² Appendix Figure A.4 shows a strong and statistically significant relationship between the two measures across; the ASHE HHI measure can explain 72% of the variation in the Lightcast HHI. Therefore, we consider the potential bias from using job vacancy data is likely to be small. Schubert et al. (2022) test the importance of workers searching beyond the boundaries of narrow local labor market definitions by constructing an innovative measure of outside options using occupational transitions recorded in US resumes. In their main IV specification, controlling for outside options only modestly reduces the elasticity of HHI with respect to wages, suggesting this is not likely to be a first order issue.

The most fundamental criticism of employer concentration as a measure of firm monopsony power is that it is itself a result of this power rather than a market primitive in its own right.²³ Existing evidence has demonstrated a positive relationship between employer concentration and other measures of monopsony power. Yeh et al. (2022) estimate plant-level wage markdowns for US manufacturing firms. Aggregating these measures up to market level, they find a positive, if weak, correlation with employer concentration. However, Azar et al. (2019) find that employer concentration is a good predictor of inelastic labor supply using job application data from an online job platform.

²²ASHE is not a perfect source of HHI data. While representative, it is only a 1% sample, which means that as a measure of concentration it will tend to be both noisy and biased upwards (a point discussed in further detail in Abel et al. (2018)). Due to sample size limitations, we can only calculate HHI at the county-SOC-1 level for each year of our Lightcast sample (2014-2019).

²³In the case of product markets, Syverson (2019) find that the relationship between market concentration and monopoly power depends, for example, on the type of market competition and product substitutability.

5.2 Empirical Specification

We seek to estimate the causal relationship between employer concentration and the prevalence of job features in a local labor market. The key concern in our setting is omitted variable bias. For example, if more productive occupations have lower market entry costs and can also provide flexibility more cheaply, this will generate a positive relationship between HHI and flexibility that does not derive from any causal impact of monopsony power on flexibility provision.

To help address this concern, all specifications will include a rich set of fixed effects to control for any systematic variation across occupations, regions, and local labor markets that might be driving the concentration-job feature relationship. Specifically, we control for local labor market (occupation x county) fixed effects ($\alpha_{o,c}$) and occupation-specific and county-specific time trends ($\delta_{o,t}$ and $\gamma_{c,t}$ respectively).

However, these market and time fixed effects will not be sufficient if there are market-level shocks that simultaneously affect changes in employer concentration and the provision of a job feature. For example, an idiosyncratic shock to productivity in a particular occupation and county could, in theory, affect both concentration and flexibility. Measurement error in HHI or the outcome variables we study could also affect the relationship between the two (for example by correlated measurement errors, or through attenuation bias).

Instrument Specification We therefore adopt an instrumental variable (IV) strategy using a Bartik-style prediction of the change in HHI based on nationwide hiring trends. Our instrument is in the vein of “granular” instrumental approach which uses changes at

the firm level to instrument for changes at the market level (Gabaix and Koijen, 2024).²⁴

Intuitively, the instrument is motivated by the idea that a firm’s local hiring is partially driven by nationwide shocks to firm demand. However, local shocks are unlikely to determine nationwide hiring.²⁵

To construct the instrument, we use the changes in nationwide hiring at multi-establishment firms to predict local changes in hiring, and then use this predicted change in local demand to construct a predicted change in HHI. To motivate this approach, note that the change in log HHI can be expressed as a function of the growth rate of vacancies at firm i , $g_{i,o,c,t}$ and the growth in vacancies across all firms in the market, $g_{i,o,c,t}$.²⁶

$$\Delta \ln(HHI_{o,c,t}) = \ln \left(\frac{\sum_i s_{i,o,c,t}^2}{\sum_i s_{i,o,c,t-1}^2} \right) = \ln \left(\frac{\sum_i s_{i,o,c,t-1}^2 \frac{g_{i,o,c,t}^2}{m_{o,c,t}^2}}{\sum_i s_{i,o,c,t-1}^2} \right) \quad (5.3)$$

In a shift-share type design, we will use the $s_{i,o,c,t-1}^2$ terms as shares and predict the $\frac{g_{i,o,c,t}^2}{m_{o,c,t}^2}$ term using firms’ nationwide hiring growth. Specifically, letting $-c$ indicate all counties other than county c , we calculate growth in the leave-one-out sum of vacancies posted by the firm for that occupation as $\hat{g}_{i,o,c,t} = \frac{v_{i,o,-c,t}}{v_{i,o,-c,t-1}}$. This expression gives us a plausibly exogenous prediction of the change in the firm’s local hiring. We then calculate predicted hiring growth for the whole market in a similar fashion: $\hat{m}_{o,c,t} = \frac{\sum_i v_{i,o,-c,t}}{\sum_i v_{i,o,-c,t-1}}$. Replacing $g_{i,o,c,t}^2$ and $m_{o,c,t}^2$ in Equation 5.3 with $\hat{g}_{i,o,c,t}$ and $\hat{m}_{o,c,t}$ gives a predicted change in $\ln(HHI_{o,c,t})$ using past hiring

²⁴Other papers using a similar approach are Schubert et al. (2022) who use it to study the impact of employer concentration on workers’ wages in the US, and Franklin et al. (2024) who use local commuting patterns to instrument for the spillover effects of a public employment program.

²⁵Note that the impact of a growing firm on HHI varies by baseline market power: in those markets where it already has significant market power growth will tend to raise HHI, while in those markets where it has limited market power growth will tend to reduce HHI.

²⁶We define $g_{i,o,c,t} = \frac{v_{i,o,c,t}}{v_{i,o,c,t-1}}$ and $m_{o,c,t} = \frac{\sum_i v_{i,o,c,t}}{\sum_i v_{i,o,c,t-1}}$.

and nationwide hiring growth.

A concern with the functional form implied by Equation 5.3 is that taking logs of a shift-share variable can lead to endogeneity even when the shifts are exogenous (Borusyak et al., 2025). Specifically, because of Jensen’s inequality, more concentrated markets will tend to have a lower IV than more competitive ones. Borusyak et al. (2025) propose having the log apply to the shift alone as a solution.²⁷ This solution is not appropriate in our setting because a firm’s predicted hiring growth ($\hat{g}$) can be very low relative to the market as a whole ($\hat{m}$). In this case, the log term becomes very negative and a poor prediction for the change in HHI . We therefore use the linear approximation of the log function which side-steps this issue while continuing to address the core concern raised by Borusyak et al. (2025). Our “linearized” IV is:

$$Z_{o,c,t} = \frac{\sum_i s_{i,o,c,t-1}^2 (\hat{g}_{i,o,c,t} - \hat{m}_{o,c,t})}{\sum_i s_{i,o,c,t-1}^2} \quad (5.4)$$

We use $Z_{o,c,t}$ as our preferred instrument for $\Delta \ln HHI_{o,c,t}$ in the structural equation:

$$\Delta f_{o,c,t} = \beta \Delta \ln HHI_{o,c,t} + \alpha_{o,c} + \delta_{o,t} + \gamma_{c,t} + \epsilon_{o,c,t} \quad (5.5)$$

where $\Delta f_{o,c,t}$ is the change in the share of vacancies with job feature f in county c and occupation o in quarter t . β is our parameter of interest. Standard errors are clustered at local labor market (county-occupation) level throughout. We weight our regressions with the size of the market (number of vacancies) in the same quarter of the previous year. As a

²⁷Specifically, Borusyak et al. (2025) propose: $\left(\left(\sum_i s_{i,o,c,t-1}^2 \ln \left(\frac{\hat{g}_{i,o,c,t}^2}{\hat{m}_{o,c,t}^2} \right) \right) / \sum_i s_{i,o,c,t-1}^2 \right)$.

result, the regression is effectively at the level of vacancy.²⁸

Relevance & Validity Our strategy relies on exogenous shocks to the nationwide hiring of thousands of firms, meaning that our identification strategy is in the vein of [Borusyak et al. \(2021\)](#), who show that shift-share instruments can be valid even if the shares used are endogenous. The instrument will be relevant if nationwide shocks to firm hiring (such as technological shocks, a new product, shifts in consumer demand or changes to the supply chain) influence the firm’s hiring at the local level. However, exogeneity will be violated if the firm’s nationwide hiring strategy is driven by shocks to a particular local labor market. While this cannot be tested empirically, our inclusion of a full set of fixed effects does control for any systematic changes at the local labor market level, for trends in occupations and over time.

Another potential concern are confounding effects from the changes in labor demand. Even though labor demand and HHI are two different concepts – HHI can both increase and decrease when labor demand goes up, depending on which part of the firm size distribution the additional hiring originates in – it is possible that a shock driving higher employer concentration might also increase local labor demand, changing the level of amenities and disamenities offered by the employer. In order to obtain an unbiased estimate of the impact of employer concentration on flexibility, we need to control for such changes in labor demand. As a result, we additionally control for quarterly relative increase in the number of vacancies in a given county and occupation.

²⁸Our specification implies that the impact of market power occurs within the contemporaneous quarter. This is consistent with our model and the high turnover of vacancies: [Bassier et al. \(2025\)](#) find that the average duration of a vacancy is 18 days. Longer delays in firms recognising changes in the labor market environment and changing their hiring practices would cause us to understate the true impact of concentration on job features.

Sample Restrictions We apply the following restrictions on our sample. First, our sample consists of multiple-establishment firms (firms posting in more than one county) with at least 10 vacancies posted within each occupation o (across all counties) in each quarter t . This brings our data more in line with the spirit of our identification strategy by reducing random noise in nationwide hiring by firms that are too small for this to be accurately measured. Second, to construct the IV for time t and occupation o we restrict our attention to vacancies posted by firms that posted at least one o vacancy at both t and $t - 1$. Third, we exclude firms with unusually high leave-one-out nationwide growth. The growth in firm hiring has a natural floor on the decrease in hiring (because firms cannot produce a negative number of vacancies) but no similar ceiling on the increase in hiring. As a result, the predicted growth $\hat{g}_{i,o,c,t}$ follows a skewed distribution with a small proportion of implausibly high growth values. We drop firms with predicted growth above 100% so that the distribution of $\hat{g}_{i,o,c,t}$ is symmetric.²⁹ We demonstrate the sensitivity of our results to these restrictions in a series of robustness checks reported in Appendix Table A.2.

Despite these restrictions, a small number of very high (or very low) IV outliers remain. We trim the IV by removing the bottom 1% of values and symmetrically the values that are above the magnitude of the bottom percentile at the top of the distribution; in practice, this bounds values of predicted change in HHI between $\pm 200\%$ quarter-on-quarter. Second, as a robustness check, we remove market-quarter observations where the IV is based on a single firm.³⁰

²⁹This restriction affects about 10% of our firm-level sample.

³⁰Each of these restrictions based on the value of the instrument corresponds to about 2% of the sample.

5.3 Results

Table 3 gives our main results and Figure 4 summarizes the main patterns graphically. In summary, changes in employer concentration do not affect the provision of all job features in the same way. A rise in concentration leads to an increase in the provision of schedule flexible jobs. However, this masks different variation across safe and risky flexible jobs. As employers' market power increases, risky flexibility rises, while the provision of safe flexibility does not significantly change. As a robustness check, we also estimate the relationship between permanent vacancies and HHI. We find that permanent vacancies decline when market concentration rises, in line with the strong existing evidence that they are a costly amenity (Datta, 2019; Mahmud et al., 2021; Bassanini et al., 2022), giving additional confidence in our approach.³¹

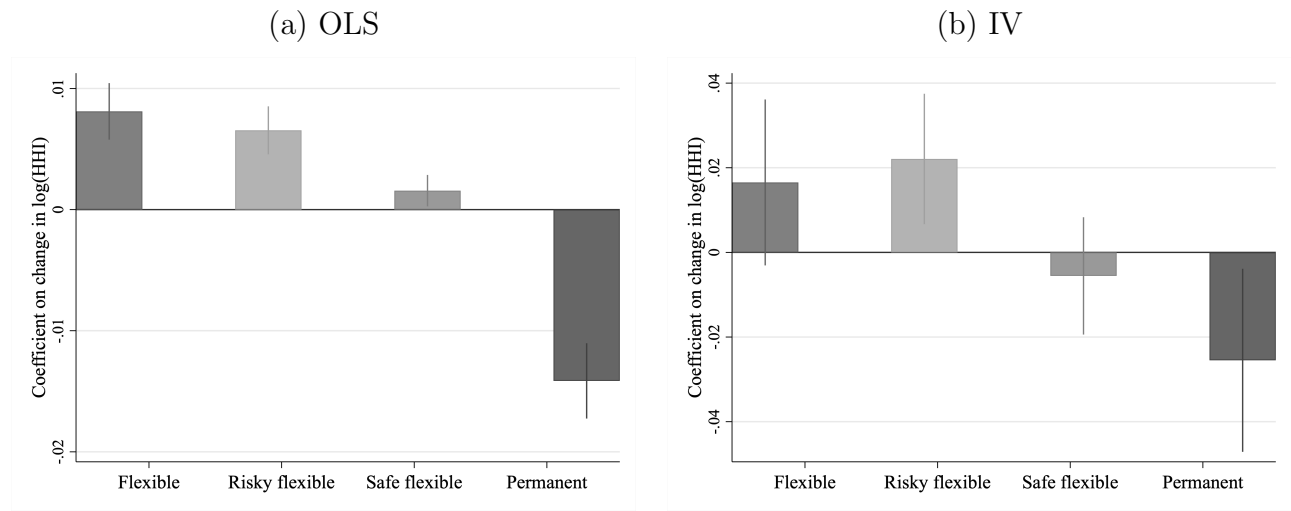
Table 3 column (1) gives the basic OLS specification including the full set of market and time fixed effects. In column (2) we add a control for the growth in number of vacancies posted in the market. In column (3) we also add controls for average wage growth and the share of vacancies with posted wage information. One reason to control for wages is to enable more precise comparisons between jobs *within* occupational cells. On the other hand, low wages are also the outcome of firm market power (e.g. Azar et al. (2020)), so wages could be a “bad control”. Our approach is to report results both with and without wage controls; we find that the controls do not change our results substantially.

Columns (4) - (6) give the equivalent results of our IV specification (Equation 5.4).

Again we begin with using only the market and time fixed effects as controls, before adding

³¹These estimates directly relate to the impact of employer concentration on advertised job characteristics; while these are likely closely connected to realised characteristics after bargaining, it is possible systematic differences between the two could alter the relationship to employer concentration.

Figure 4: Flexible Vacancies and Employer Concentration



Notes: The relationship between the change in employer concentration ($\log(\text{HHI})$) and the changes in the share of vacancies offering the given job features, as estimated by the FE regression (panel (a)) and the IV regression (panel (b)). Each bar plots the size of the coefficient from estimating equation 5.5. Positive value implies the job feature is a profitable disamenity; negative value implies it is a costly amenity; and values close to 0 imply a profitable amenity. The regression includes fixed effects at county x SOC-3 level, county x year-quarter level, and SOC-3 x year-quarter level. The 95% confidence intervals are calculated from standard errors clustered at county-occupation level.

in controls for growth in number of vacancies posted and wages. The chief message from Table 3 is that our results remain robust throughout. Risky flexibility is more prevalent in more concentrated labor markets, with a positive and statistically significant coefficient for all specifications. The impact is considerably larger in the IV specifications than the OLS ones. This may reflect local productivity shocks. If rising concentration is driven by higher local productivity, the resulting productivity gains should improve job quality (reducing risky flexibility), partially offsetting the direct negative effect of greater concentration. The effect of HHI on safe flexible vacancies is much smaller than the effect on risky vacancies in the OLS fixed effect specifications, and is not statistically significant from zero in the IV specifications. Permanent vacancies become less common as concentration increases.

We subject our main IV results to a battery of additional checks. First, we begin with several changes to the sample and specification in Table A.2. We re-run our results with two different sample restrictions: excluding markets whose IV is based on a single multi-establishment firm; and including firms with very high growth based on their leave-one-out nationwide hiring. We also use the alternative IV described in equation 5.3. Finally, it is possible that there are spatial correlations in the error terms, leading to an underestimate of standard errors; as a simple check, we repeat our core regressions but at a higher level of spatial aggregation - using NUTS2 regions (of which there are 40), rather than counties (of which there are 206). These variations all deliver similar results: the impact of HHI on risky flexible vacancies remains statistically significantly positive; the impact on safe flexible vacancies remains insignificant and close to zero; and the impact on permanent vacancies remains negative, though not always statistically significant.

To further check that our results are not being driven by a small number of outliers, the

left hand side of Appendix Figure A.5 visualises the OLS fixed effects results by showing bin-scatters of job characteristics and the change in HHI after having residualised on α_{oc} , δ_{ot} , and γ_{ct} . In the right hand figures we also condition on the residual from the IV first stage, visualising the IV results (in the spirit of a control function approach). Across outcomes, the plots show a smooth and monotonic relationship between employer concentration and job features and provide reassurance that the estimated effects are not driven by outliers or thin local labor markets. While the IV-based plots are naturally noisier, the qualitative patterns remain consistent.

Another potential concern is that our measured effect of greater market concentration on flexible vacancies is driven by broader trends in the characteristics of vacancies rather than by the impact of changing market power per se. To test for such compositional changes, we report in Table A.3 the results of our core regression where the outcome variable is the number of skills listed in the vacancy text, estimated both for our baseline sample and excluding single firm markets. We find no significant effect of HHI on listed skills. This suggests that our results on the impact of greater HHI on flexibility is driven by firms using their market power to reduce a given job’s quality, rather than by shifts in the composition of advertised jobs.

As a final test of our approach, we replicate the standard result that increases in HHI reduce wages.³² Table A.3 shows the effect of HHI on two measures of wages - log wage, and a dummy indicating that the vacancy’s wage is below the average for the occupation. We

³²This is a somewhat challenging exercise as 37% of vacancies do not list a wage. Ideally we would measure concentration with the Lightcast Data and wages with another data source, as in Azar et al. (2020); unfortunately there is no separate UK data source which has nearly a large enough sample to measure wages over time at the occupation x county level. We consider the previously discussed impact of HHI on permanent job status, which we observe for all vacancies, to be a cleaner check on our method.

again show this for the full sample and excluding single firm markets. In all cases we find effects in the expected direction, with p-values between 0.07 and 0.11.

Discussion Our results show that the interaction of salary type and flexible scheduling pick out different types of jobs. In addition to the wage and skill content of risky and safe flexible positions being systematically different (in ways that go beyond the differences in salaried versus non-salaried positions – see Table 2), they are also differentially affected by changes in employer market power. Interpreted through the lens of our model, our estimates imply that risky flexibility is a disamenity that increases firms’ profits while safe flexibility is an amenity, possibly a profitable one.

Our concept of risky flexibility is conceptually distinct from that of employer-controlled schedule flexibility.³³ However, the two concepts are clearly related and interact in determining worker utility: a worker’s exposure to earnings risk will be greater in jobs where they do not receive a fixed salary *and* do not control their schedule. To analyze the role that employer control plays in driving the relationship between risky flexible jobs and employer concentration, we interact the change in $\log(HHI)$ in our baseline OLS specification with the proportion of employer-controlled schedule flexible jobs at the 1-digit occupation level measured from the Understanding Society survey data:

$$\Delta f_{o,c,t} = \beta \Delta \log HHI_{o,c,t} + \rho \Delta \log HHI_{o,c,t} \times Employer_control_O + \alpha_{o,c} + \delta_{o,t} + \gamma_{c,t} + \epsilon_{o,c,t} \quad (5.6)$$

where $Employer_control_O$ is the proportion of employer-controlled schedule flexible contracts (both with an without a minimum number of guaranteed hours) out of all flexible contracts in the 1-digit SOC code encompassing occupation o .

Figure 5 shows the marginal effect of a change in employer concentration on risky flex-

³³We cannot measure employer control of schedules from the vacancy text. See Section 2.

Table 3: The relationship between employer concentration and flexibility

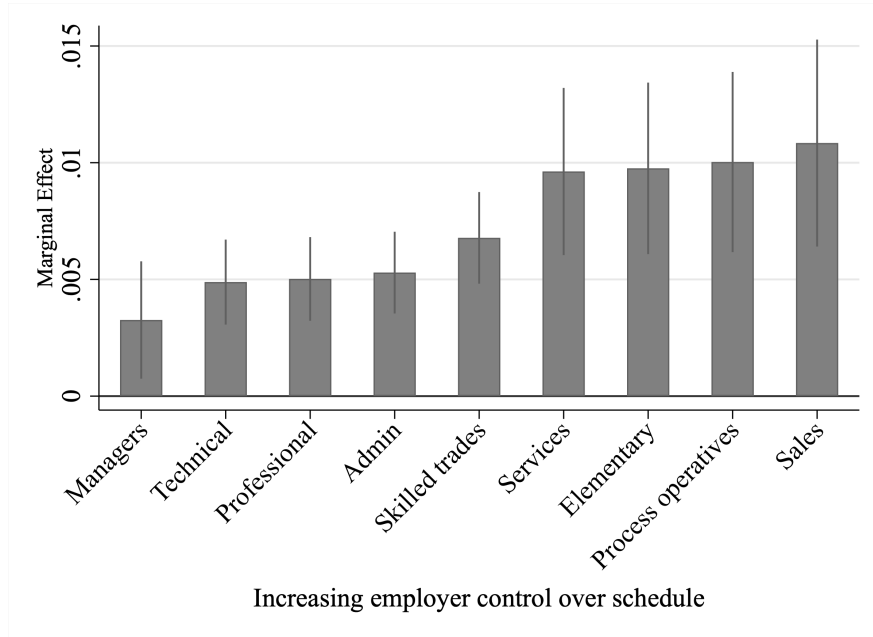
	FE			IV		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Flexible Vacancies</i>						
$\Delta \log(\text{HHI})$	0.00811*** (0.00119)	0.00899*** (0.00133)	0.00899*** (0.00133)	0.0165* (0.01000)	0.0161 (0.0106)	0.0162 (0.0106)
Growth in hiring		0.00146** (0.000640)	0.00145** (0.000641)		0.000741 (0.00132)	0.000714 (0.00132)
$\Delta \log(\text{wage})$			-0.00384* (0.00208)			-0.00376* (0.00221)
<i>Panel B: Risky Flexible Vacancies</i>						
$\Delta \log(\text{HHI})$	0.00655*** (0.00101)	0.00658*** (0.00114)	0.00658*** (0.00114)	0.0221*** (0.00786)	0.0216*** (0.00836)	0.0217*** (0.00836)
Growth in hiring		0.00251*** (0.000617)	0.00250*** (0.000618)		0.000887 (0.00111)	0.000867 (0.00111)
$\Delta \log(\text{wage})$			-0.00402** (0.00174)			-0.00393** (0.00197)
<i>Panel C: Safe Flexible Vacancies</i>						
$\Delta \log(\text{HHI})$	0.00156** (0.000661)	0.00242*** (0.000735)	0.00241*** (0.000734)	-0.00555 (0.00707)	-0.00547 (0.00753)	-0.00546 (0.00753)
Growth in hiring		-0.00105*** (0.000397)	-0.00105*** (0.000397)		-0.000145 (0.000911)	-0.000153 (0.000910)
$\Delta \log(\text{wage})$			0.000185 (0.00131)			0.000165 (0.00136)
<i>Panel D: Permanent Vacancies</i>						
$\Delta \log(\text{HHI})$	-0.0141*** (0.00159)	-0.0134*** (0.00178)	-0.0133*** (0.00174)	-0.0255** (0.0110)	-0.0252** (0.0118)	-0.0266** (0.0118)
Growth in hiring		-0.00195** (0.000883)	-0.00178** (0.000874)		-0.000595 (0.00160)	-0.000251 (0.00158)
$\Delta \log(\text{wage})$			0.0379*** (0.00232)			0.0382*** (0.00233)
N	217790	116784	116784	114583	114583	114583
1st-stage F				168.0	162.1	162.3
Safe vs risky p-value	0.0000464	0.00267	0.00266	0.0129	0.0223	0.0217

Notes: The relationship between the change in employer concentration ($\log(\text{HHI})$) and the changes in the share of vacancies offering the given job features. The table reports coefficients from estimating equation 5.5, using 5.4 as an instrument for the change in $\log(\text{HHI})$. Positive value implies the job feature is a profitable disamenity; negative value implies it is a costly amenity; and values close to 0 imply a profitable amenity. All regressions include the full set of fixed effects: county x SOC3, year-quarter x SOC3, and year-quarter x county. We report Kleibergen-Paap Wald F statistic for the 1st stage of the IV regressions. “Safe vs risky p value” presents the p-values that the coefficients on $\Delta \log(\text{HHI})$ for safe flexibility and risky flexibility are equal to each other. Standard errors clustered at county-occupation level in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

ibility by occupation group given the prevalence of employer-controlled flexibility in that occupation. The estimated interaction effect between HHI and employer-controlled schedule is positive and statistically significant (0.0135, p-value of 0.0124). The positive relationship between employer concentration and risky flexibility is much stronger for those occupations in which employer-controlled flexibility is most prevalent. This is reasonable: in those occupations where employees have little control over their schedule and their earnings fluctuate with variation in the number and timing of hours, flexibility is a clear disamenity for workers. In occupations where workers can control their schedules, risky flexibility reduces workers' utility by less. These results suggest that our approach could be harnessed to identify whether the flexibility experienced by different groups of workers is an amenity or disamenity in data sets that include information on flexibility and salary type, even when measures of schedule control are missing.

Figure 5: Occupation-Specific Results: Risky Flexibility & Employer Schedule Control



Notes: Figure plots the marginal effect of $\Delta \log(HHI)$ on the share of vacancies offering risky flexibility across occupations, ordered by the observed prevalence of employer-controlled flexibility in that occupation. The coefficients are based on a equation 5.6 with $\hat{\beta} = -0.0006$ (0.0026) and $\hat{\rho} = 0.0135$ (0.0054). The regression includes fixed effects at county x SOC-3 level, county x year-quarter level, and SOC-3 x year-quarter level. The 95% confidence intervals are calculated from standard errors clustered at county-occupation level. The coefficients are ordered from left to right by increasing levels of employer-controlled flexibility as measured by the proportion of flexible contracts that are employer-controlled in Understanding Society survey data.

6 Conclusion

This paper shows that schedule flexibility is not a single, uniformly valued job attribute. Using new measures of flexible work arrangements derived from the text of online job vacancies, we document substantial heterogeneity in the type of flexibility offered in the labor market. In particular, we distinguish between safe flexibility, which preserves income stability through salaried contracts, and risky flexibility, which exposes workers to earnings volatility through unsalaried arrangements. We show that risky flexibility is concentrated in low-wage, low-skill occupations and in less competitive labor markets, while safe flexibility is more prevalent in higher-paying, higher-skilled jobs. Even within occupations and local labor markets, risky flexible vacancies are associated with lower wages and lower skill requirements than both safe flexible and fixed-schedule jobs.

To interpret these patterns, we develop a simple model of wage and job-feature provision under employer market power and test its predictions empirically. We find that higher employer concentration causes an increase in risky flexible jobs, while the prevalence of safe flexibility is largely unaffected. Interpreted through the lens of the model, these results suggest that risky flexibility functions as a profitable disamenity, whereas safe flexibility is an amenity. Our findings imply that focusing on wages alone understates the welfare consequences of monopsony power, as labor market concentration also reshapes the non-wage attributes of jobs. More broadly, the results caution against treating schedule flexibility as an unambiguously beneficial feature and highlight the importance of distinguishing between forms of flexibility that insure workers against risk and those that shift risk onto them.

Our results suggest several further avenues for research. First, in our data we cannot observe realized outcomes once a worker is matched to a vacancy. Combining the measurement approach developed in this paper, with matched employer-employee data would facilitate a detailed analysis of career trajectories in flexible positions, something we know little about. Second, we found systematic differences in the skill content of risky flexible and safe flexible jobs; risky flexible vacancies are associated with the fewest number of skills demanded,

less than both safe flexible and non-flexible vacancies. Understanding the reason for these skill differences is important for understanding the long term consequences for employment progression, worker welfare, and productivity of work in flexible jobs, especially if these positions continue to grow in popularity. Finally, the machinery developed in this paper can be applied to other important non-wage amenities, including health insurance provision and on-the-job training.

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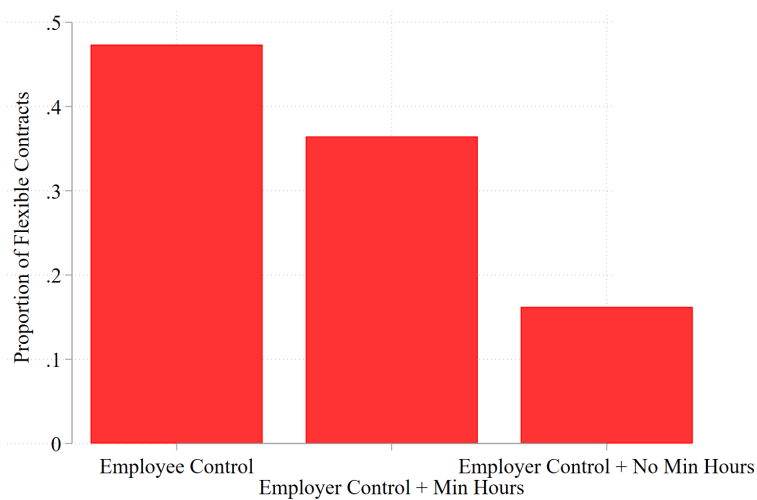
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Online Appendix for “Firm Concentration & Job Design: The Case
of Schedule Flexible Work Arrangements”

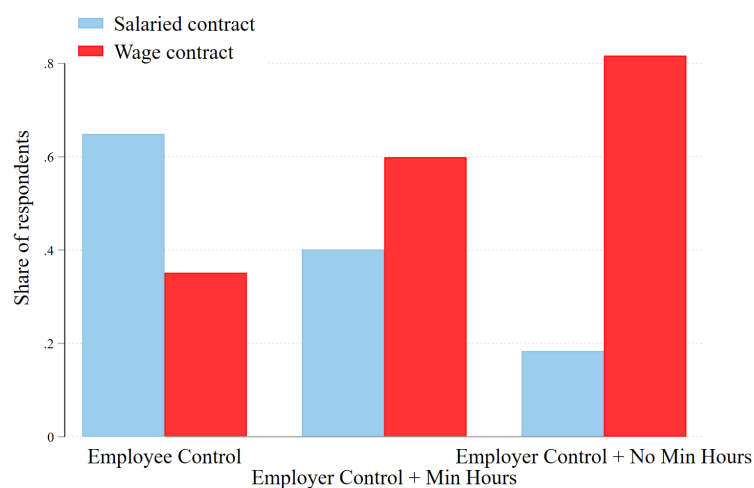
A Additional Figures and Tables

Figure A.1: Schedule Control & Remuneration Type

(a) Schedule control among workers without fixed hours

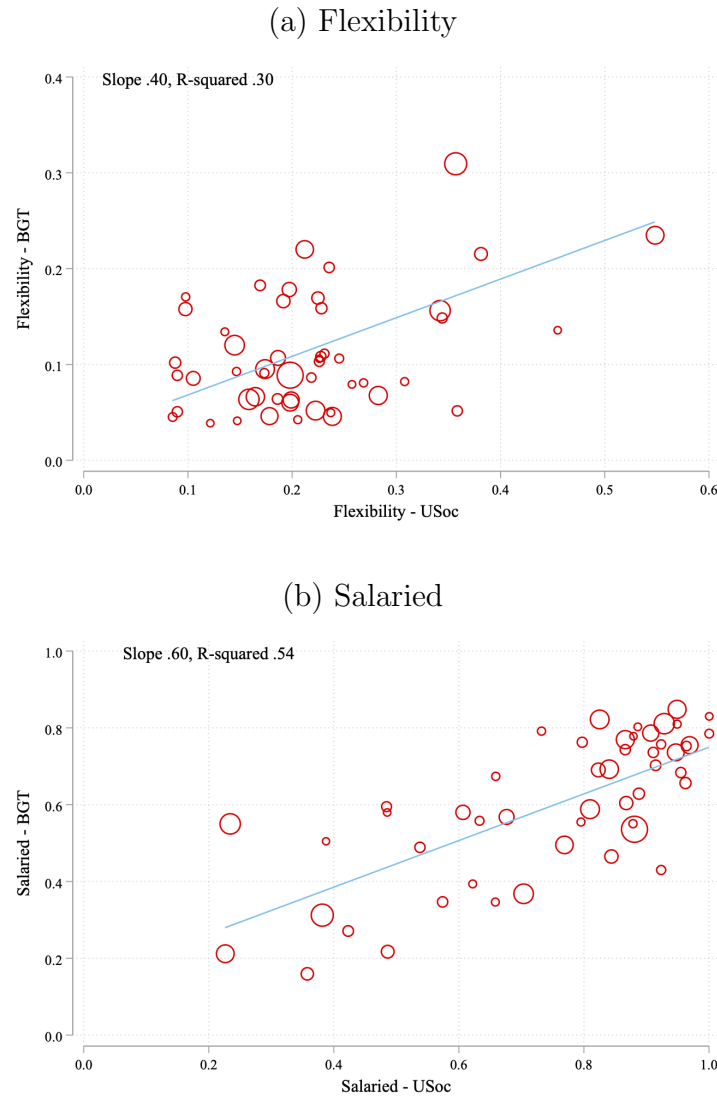


(b) Remuneration type conditional on schedule control



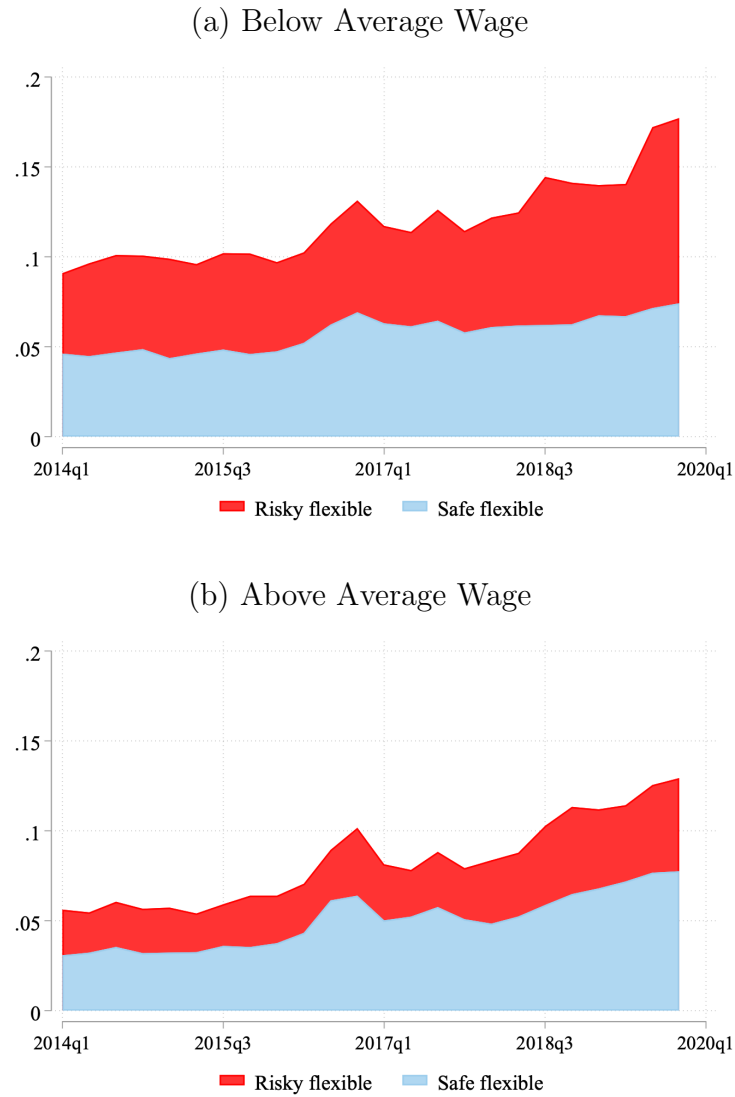
Notes: Understanding Society Covid-19 panel data. Sample restricted to those in employment in January 2020 in a job without a fixed number of hours and without missing values on answers to job flexibility and remuneration type of job in January 2020.

Figure A.2: Correlation Between Measures in Understanding Society & Lightcast



Notes: Scatter plot of the average proportion of (a) flexible contracts and (b) salaried contracts across 3-digit occupations codes in Understanding Society Covid-19 panel data (x-axis) and in Lightcast (y-axis). Sample in USoc restricted to those in employment in January 2020 and without missing values on answers to job flexibility and remuneration type of job in January 2020. Size of the scatter point is proportional to the relative size of a 3-digit SOC code in the USoc data.

Figure A.3: Change in Schedule Flexibility Over Time by Wage Group



Notes: Lightcast data from 2014-2019. The sample is restricted to job adverts with an occupation code, wage, and county. Wages deflated to the 2019 price level. Vacancies split according to whether record an above or below mean wage.

Figure A.4: Correlation Between Measures in ASHE datasets & Lightcast

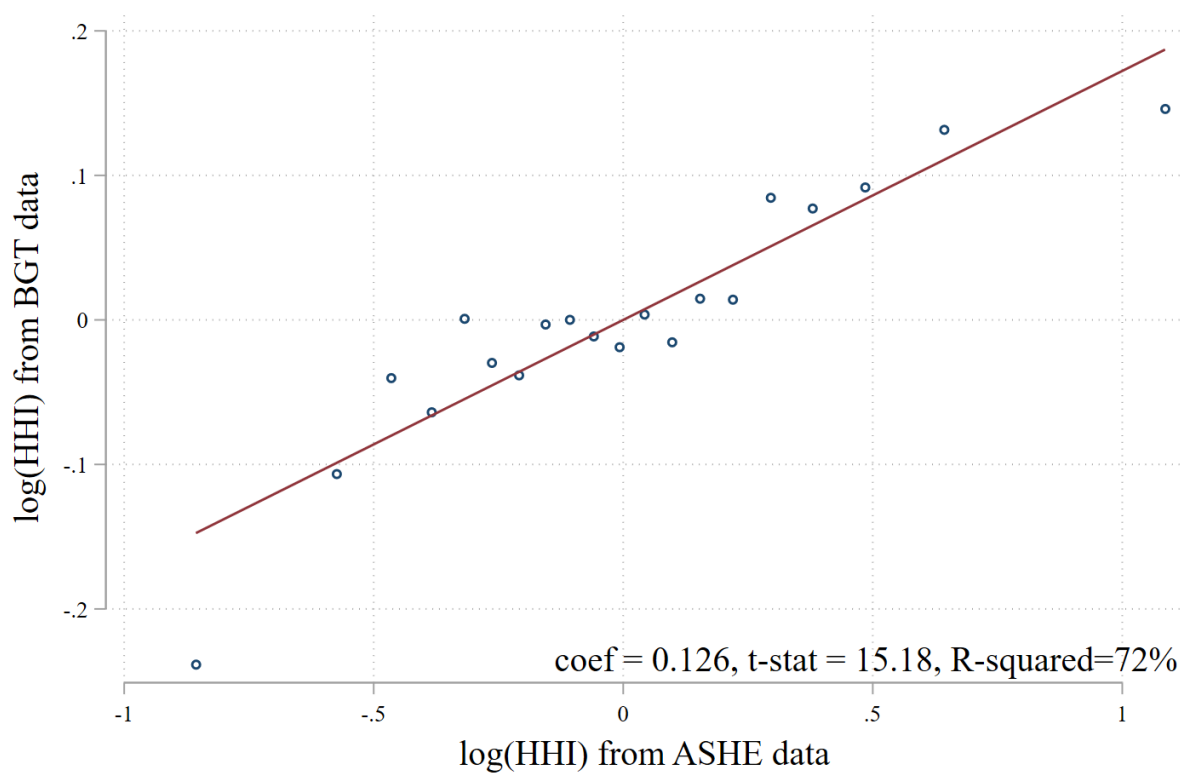
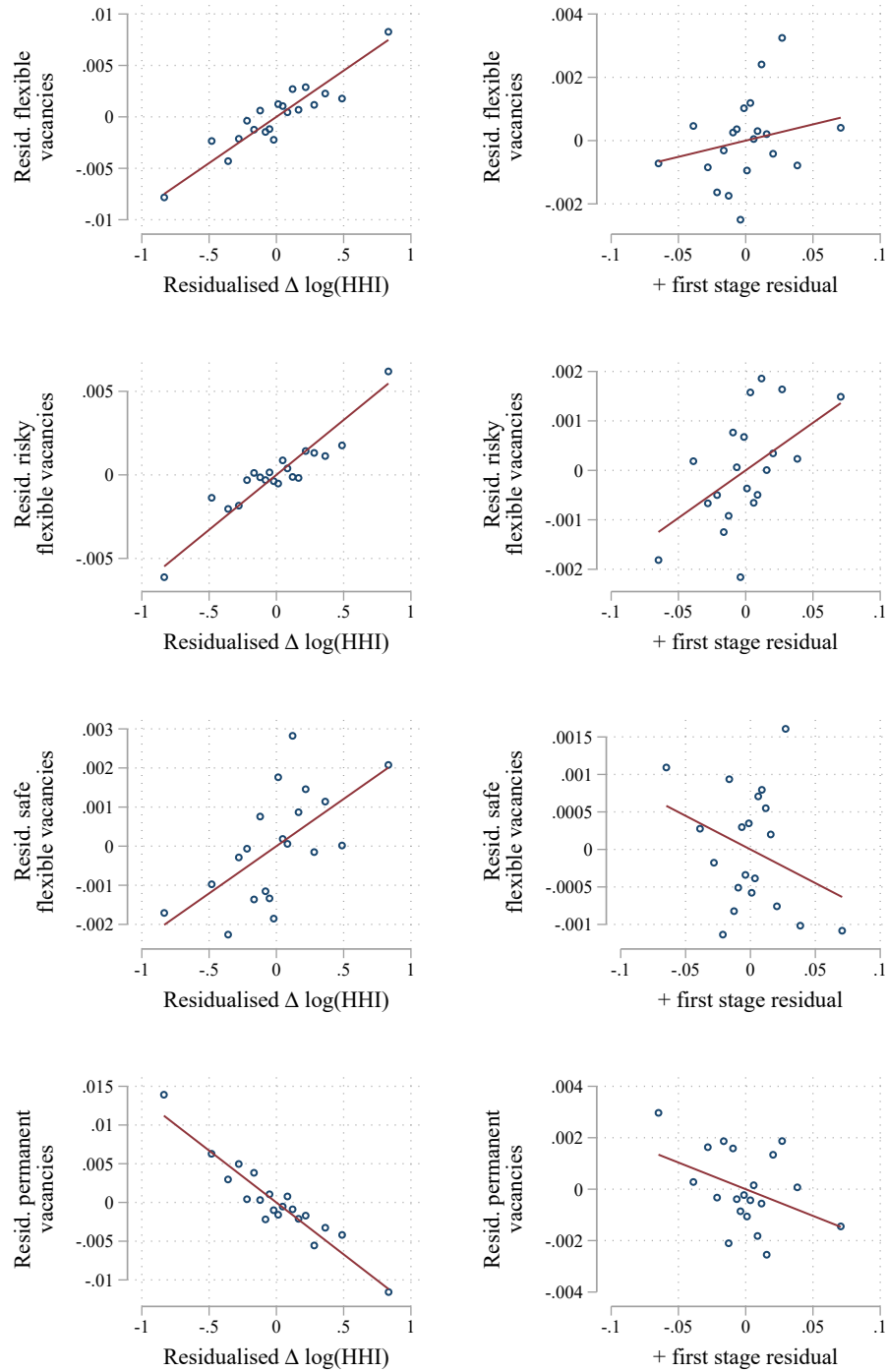


Figure A.5: Relationships between concentration and job amenities



Notes: The figures on the left hand side show the relationship between the change in log HHI and a job feature, after residualising both against growth in labor demand and the α_{oc} , δ_{ot} , and γ_{ct} fixed effects from equation 5.5. This visually shows the relationship that drives the results from our baseline fixed effects regressions. In the right hand side figures we first run the first stage IV regression and extract the residual. We then residualise the change in log HHI and the job feature against the same marked demand control and fixed effects as before, as well as the first stage residual. This is a control function approach and approximates the relationship that drives the results from our IV regressions.

Table A.1: Descriptive Statistics: Employer Concentration & Local labor Market Size

	Mean	St. Dev.	10th pct.	90th pct.
<i>Number of vacancies per</i>				
Market	691	4279	8	1371
Firm	26	338	1	16
Firm x Occupation	7.9	81	1	10
<i>Employer concentration</i>				
HHI	2740	2921	327	10000
HHI (vacancy-weighted)	615	1102	35	1449
Firm hiring share	.46	1.5	.015	1
Share of single-establishment firms	.24			
<i>Total sample size</i>				
Market	17933			
Firm	482065			
Firm x Occupation	1571369			

Notes: The first panel gives the number of vacancies within local labor market, firm, and per firm-occupation. The second panel describes employer concentration: the average HHI index of a local labor market and the largest firm-occupation importance weight in the dataset. Note that we define the HHI so that it falls between 1 (perfect competition) and 10,000 (single monopsony employer). The third panel summarizes the number of observations as the number of firm-occupations combinations. All statistics are calculated for pooled data with the exception of the middle panel (Employer concentration), where the statistics are summarized over year-quarters.

Table A.2: The relationship between employer concentration and flexibility: robustness checks

	(1) Baseline IV	(2) Excl. single-firm markets	(3) Incl. high firm growth	(4) Alternative IV	(5) NUTS2
<i>Panel A: Flexible Vacancies</i>					
$\Delta \log(\text{HHI})$	0.0161 (0.0106)	0.0209** (0.0105)	0.0118 (0.00836)	0.0126 (0.0199)	0.0161 (0.0109)
Growth in hiring	0.000741 (0.00132)	0.000512 (0.00231)	0.00136 (0.00104)	0.00171 (0.00197)	0.000741 (0.00136)
<i>Panel B: Risky Flexible Vacancies</i>					
$\Delta \log(\text{HHI})$	0.0216*** (0.00836)	0.0198** (0.00814)	0.0142** (0.00630)	0.0346** (0.0159)	0.0216** (0.00870)
Growth in hiring	0.000887 (0.00111)	0.00120 (0.00199)	0.00184** (0.000867)	0.000287 (0.00162)	0.000887 (0.00115)
<i>Panel C: Safe Flexible Vacancies</i>					
$\Delta \log(\text{HHI})$	-0.00547 (0.00753)	0.00114 (0.00737)	-0.00240 (0.00580)	-0.0221 (0.0141)	-0.00547 (0.00771)
Growth in hiring	-0.000145 (0.000911)	-0.000687 (0.00152)	-0.000481 (0.000682)	0.00142 (0.00140)	-0.000145 (0.000936)
<i>Panel D: Permanent Vacancies</i>					
$\Delta \log(\text{HHI})$	-0.0252** (0.0118)	-0.0188 (0.0118)	-0.0171* (0.00912)	-0.0116 (0.0231)	-0.0252** (0.0118)
Growth in hiring	-0.000595 (0.00160)	-0.000441 (0.00300)	-0.000814 (0.00119)	-0.00240 (0.00229)	-0.000595 (0.00162)
N	114583	74503	110935	96021	114583
1st-stage F	162.1	157.5	219.6	43.40	153.4

Notes: The relationship between the change in employer concentration ($\log(\text{HHI})$) and the changes in the share of vacancies offering the given job features. The table reports coefficients from estimating equation 5.5, using 5.4 (columns (1)-(3) and (5)) and 5.3 (column (4)) as an instrument for the change in $\log(\text{HHI})$. Positive value implies the job feature is a profitable disamenity; negative value implies it is a costly amenity; and values close to 0 imply a profitable amenity. Columns (1) reprints our baseline IV estimate. Column (2) excludes markets where IV is predicted by a single firm. The IV in column (3) includes firms with very high vacancy growth. Column (4) uses an alternative definition of IV based on equation 5.3. Column (5) uses an alternative spatial structure (NUTS2) for the definition of local labor market. All regressions include the full set of fixed effects: county x SOC3, year-quarter x SOC3, and year-quarter x county, and control for the growth in local hiring. We report Kleibergen-Paap Wald F statistic for the 1st stage of the IV regressions. Standard errors clustered at county-occupation level in parentheses. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$

Table A.3: The relationship between employer concentration and number of skills and wages

	No. Skills		Low Wage		Ln Wage	
	(1) All	(2) Ex. Single	(3) All	(4) Ex. Single	(5) All	(6) Ex. Single
$\Delta \log(\text{HHI})$	-0.0228 (0.101)	0.0235 (0.0986)	0.0186* (0.0104)	0.0190* (0.0104)	-0.0178 (0.0110)	-0.0178* (0.0108)
Growth in hiring	-0.0311** (0.0140)	-0.0525** (0.0245)	-0.00173 (0.00129)	-0.00400* (0.00239)	0.00490*** (0.00127)	0.00710*** (0.00248)
$\Delta \log(\text{wage})$	-0.193*** (0.0197)	-0.187*** (0.0250)				
N	114583	74503	112551	74261	111275	74108
Mean Dep. Var						
1st-stage F	162.3	157.8	161.1	153.6	161.1	153.8

Notes: Estimating the relationship between the change in employer concentration ($\log(\text{HHI})$) and the changes in the average number of skills and log wages at the market level. All regressions include the full set of fixed effects: county x SOC3, year-quarter x SOC3, and year-quarter x county. We report Kleibergen-Paap Wald F statistic for the 1st stage of the IV regressions. Number of skills measured as the average number of skill requirements listed in a job ad at the market level. ‘Low wage’ is the proportion of vacancies at the market level below average real wages for that occupation. Ln wage is measured as the logarithm of the average market wage (trimmed to exclude the top 1% of observations given outliers). Columns (2), (4) and (6) exclude single firm markets. Standard errors clustered at county-occupation level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ Sample restricted to vacancies with information on wage, county, and occupation.

B Proofs for the theoretical model

B.1 Model solution

The first-order conditions of the optimisation problem in (4.2) are:

$$\text{FOC}_w : \quad \frac{\partial l}{\partial U} U'(w)[y - w + \delta(f)] - l(U(w, f)) = 0 \quad (\text{B.1})$$

$$\text{FOC}_f : \quad \frac{\partial l}{\partial U} U'(f)[y - w + \delta(f)] + l(U(w, f))\delta'(f) = 0. \quad (\text{B.2})$$

where the first term in each condition captures the general-equilibrium feedback between firm's labor demand and equilibrium utility.

Simplifying and writing in elasticities yields:

$$\text{FOC}_w : \quad w = [y + \delta(f)] \frac{\epsilon \eta_w}{\epsilon \eta_w + \lambda} \quad (\text{B.3})$$

$$\text{FOC}_f : \quad y - w = -\frac{\lambda}{\epsilon \eta_f} f \delta'(f) - \delta(f) \quad (\text{B.4})$$

where $\epsilon = \frac{UL'(U)}{L}$ is the elasticity of labor supply with respect to market utility, $\eta_w = \frac{wU'(w)}{U}$ is the elasticity of utility with respect to wages, and $\eta_f = \frac{fU'(f)}{U}$ is the elasticity of utility with respect to amenities. The ratio l/L appearing in the derivation equals $1/N$ under firm homogeneity firms; we denote this by λ .³⁴ Larger λ therefore corresponds to stronger employer power and greater internalization of the market-wide feedback. On the other end, as $\lambda \rightarrow 0$ the model collapses to the competitive benchmark.

To find optimal w^*, f^* , we substitute the expression for wage from FOC(w) into FOC(f). This yields the following implicit solution for f^* :

$$y = -\frac{\epsilon \eta_w + \lambda}{\epsilon \eta_f} f^* \delta'(f^*) - \delta(f^*) \quad (\text{B.5})$$

³⁴If all firms are of equal size, each will employ $1/N$ of the market, where N is the number of firms. The measure of market concentration λ , defined as HHI , is then $\lambda = \sum (1/N)^2 = 1/N$.

B.2 Proof of Proposition 1

Proof. We prove Proposition 1 by total differentiation of the implicit solution for f , expression (B.5). Differentiating this expression with respect to λ :

$$\begin{aligned}\frac{\partial y}{\partial \lambda} &= -\frac{1}{\epsilon \eta_f} f^* \delta'(f^*) - \frac{\epsilon \eta_w + \lambda}{\epsilon \eta_f} \left[\frac{df^*}{d\lambda} \delta'(f^*) + \frac{df^*}{d\lambda} f^* \delta''(f^*) \right] - \delta'(f^*) \frac{df^*}{d\lambda} \\ 0 &= \frac{df^*}{d\lambda} \left[\frac{\epsilon \eta_w + \lambda}{\epsilon \eta_f} (\delta'(f^*) + f^* \delta''(f^*)) + \delta'(f^*) \right] + \frac{1}{\epsilon \eta_f} f^* \delta'(f^*) \\ \frac{df^*}{d\lambda} &= -\frac{f^* \delta'(f^*)}{\delta'(f^*)(\epsilon \eta_w + \epsilon \eta_f + \lambda) + f^* \delta''(f^*)}\end{aligned}\tag{B.6}$$

The last equation describes the equilibrium relationship between market concentration and the optimal provision of job feature f .

For costly amenities, the numerator is negative and the denominator, a sum of two negative terms, is also negative. This makes the negative of this expression also negative: firms offer fewer costly amenities when their monopsony power is larger.

In the case of profitable disamenities, the numerator is positive and the denominator is negative as long as the $\delta(f)$ function displays sufficiently strong decreasing returns to scale (or workers' utility is highly elastic with respect to job features f). This makes the expression positive: firms increase the provision of disamenities when their monopsony power increases.

The expression is negative for profitable amenity because the interior solution for f^* is found on the downward-sloping portion of $\delta(f)$. However, given that f^* is higher under profitable amenity than costly amenity, the response to rising λ will be smaller than for costly amenities and tend to 0. To see this formally, denote f_P^* and f_C^* as optimal profitable amenity than costly amenity, respectively. Assume that f_P^* is in a similar position of the downward-sloping portion of $\delta(f)$ as f_C^* , so that $\delta'(f_C^*) = \delta'(f_P^*)$ and $\delta''(f_C^*) = \delta''(f_P^*)$. Then it will be true that the adjustment in profitable amenity is larger (smaller in magnitude) if

the following is true:

$$\begin{aligned}
\frac{df_P^*}{d\lambda} &= -\frac{f_P^* \delta'(f_P^*)}{\delta'(f_P^*)(\epsilon\eta_w + \epsilon\eta_f + \lambda) + f_P^* \delta''(f_P^*)} > -\frac{f_C^* \delta'(f_P^*)}{\delta'(f_P^*)(\epsilon\eta_w + \epsilon\eta_f + \lambda) + f_C^* \delta''(f_P^*)} = \frac{df_C^*}{d\lambda} \\
&\frac{f_P^*}{\delta'(f_P^*)(\epsilon\eta_w + \epsilon\eta_f + \lambda) + f_P^* \delta''(f_P^*)} > \frac{f_C^*}{\delta'(f_P^*)(\epsilon\eta_w + \epsilon\eta_f + \lambda) + f_C^* \delta''(f_P^*)} \\
&f_P^* \delta'(f_P^*)(\epsilon\eta_w + \epsilon\eta_f + \lambda) < f_C^* \delta'(f_P^*)(\epsilon\eta_w + \epsilon\eta_f + \lambda) \\
&f_P^* > f_C^*
\end{aligned}$$

Optimal provision of profitable amenities will be always higher than the optimal provision of costly amenities, so this condition holds. More broadly, if f_P^* lies relatively closer to b , the first and second derivatives of $\delta(f)$ will be close to 0. As $\frac{df_C^*}{d\lambda}$ is always negative, $\frac{df_C^*}{d\lambda} < \frac{df_P^*}{d\lambda}$, i.e. the response of f_P^* to changes in λ will be smaller than the changes in f_C^* .

■

C Representativeness of Lightcast data

Table A.4: Summary Statistics, 2014-2019

	Lightcast data	ONS data
Number of vacancies (millions)	46	55.4
Share salaried (%)	74.8%	60.2%
Share permanent (%)	66.3%	92.1%
Share with wage info (%)	63.1%	
< £9	22.1%	24%
£9 – £15	39.9%	36.7%
£15 – £20	17.3%	16.1%
> £20	20.8%	23.2%

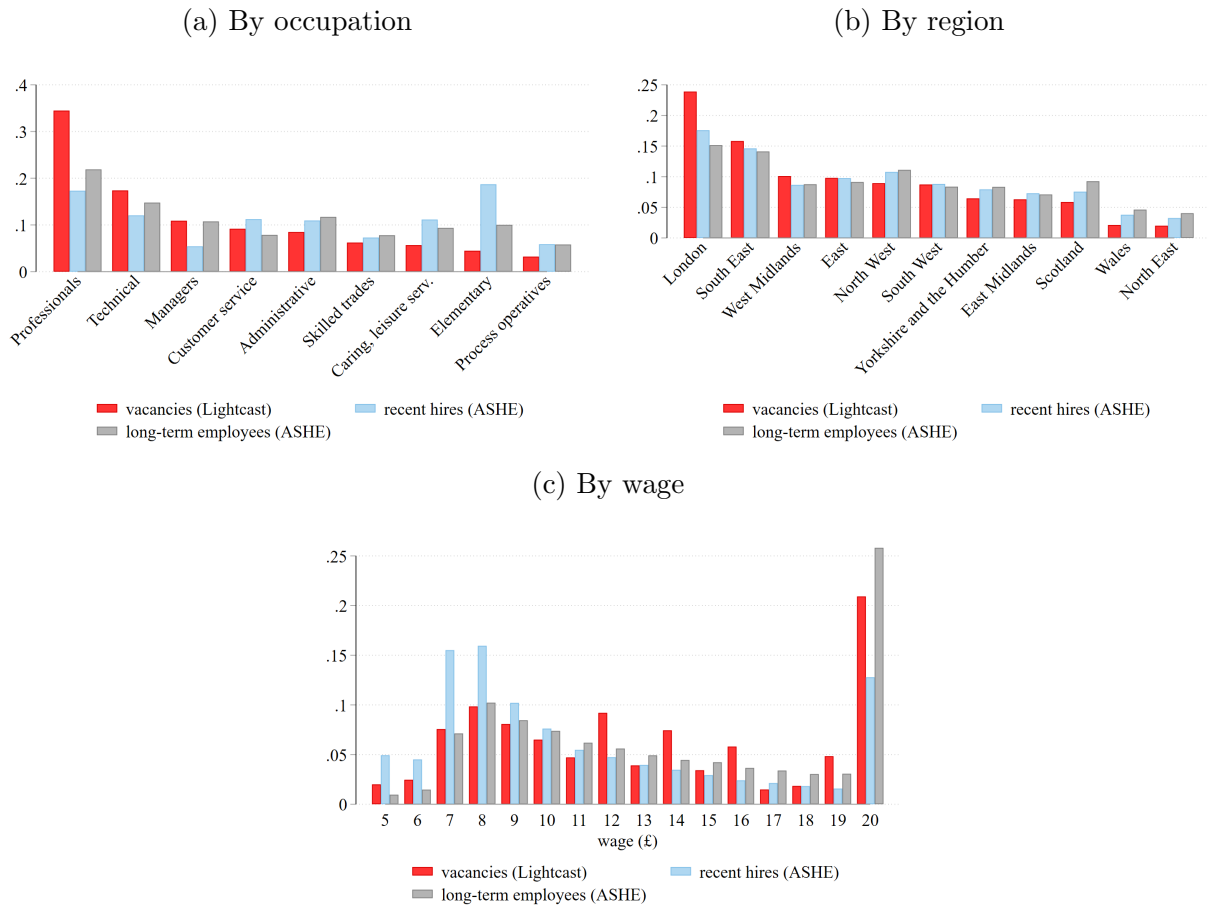
Notes: ONS data on the number of vacancies comes from the Vacancy Survey. The ONS data in the rest of the table comes from the Annual Survey of Hours and Earnings (and therefore measures the stock of employees). Data is pooled over the 2014-2019 period.

Table A.5: Descriptive Statistics for Vacancies with and without Wage Information

	wage missing		difference
	wage missing	wage included	-
N. of skills	5.0558 (4.6999)	5.3602 (5.5735)	-0.3045 (0.0015)
Permanent contract (%)	0.5799 (0.4936)	0.4308 (0.4952)	0.1491 (0.0002)
Schedule flexible (%)	0.2244 (0.4172)	0.1755 (0.3804)	0.0488 (0.0001)
Managers (%)	0.1080 (0.3104)	0.1102 (0.3131)	-0.0022 (0.0001)
Professionals (%)	0.3414 (0.4742)	0.3495 (0.4768)	-0.0081 (0.0001)
Technical (%)	0.1747 (0.3797)	0.1722 (0.3776)	0.0025 (0.0001)
Administrative (%)	0.0893 (0.2852)	0.0785 (0.2689)	0.0108 (0.0001)
Skilled trades (%)	0.0637 (0.2442)	0.0596 (0.2368)	0.0040 (0.0001)
Caring, leisure serv. (%)	0.0561 (0.2302)	0.0575 (0.2328)	-0.0014 (0.0001)
Customer service (%)	0.0910 (0.2876)	0.0937 (0.2915)	-0.0028 (0.0001)
Process operatives (%)	0.0344 (0.1822)	0.0282 (0.1654)	0.0062 (0.0001)
Elementary (%)	0.0414 (0.1993)	0.0506 (0.2192)	-0.0092 (0.0001)

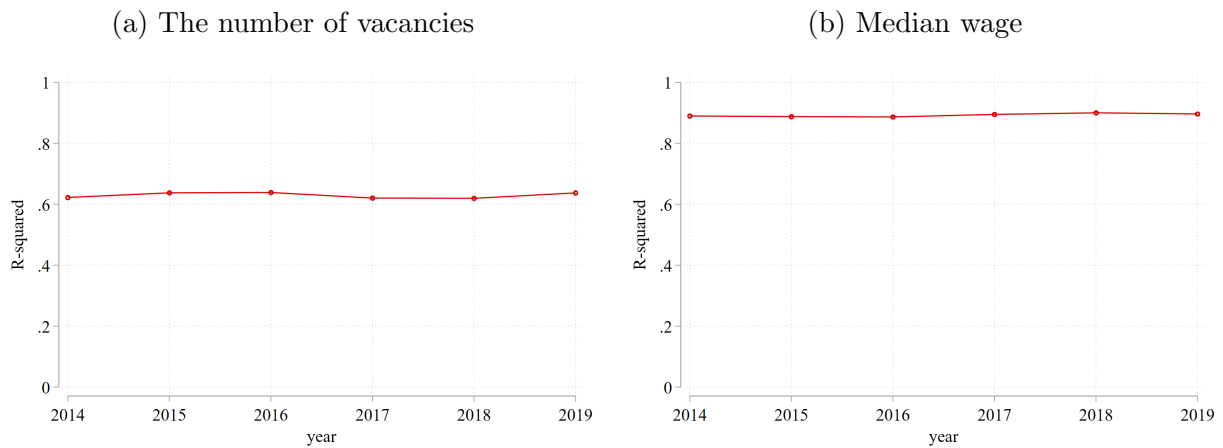
Notes: The number of required skills, the share of permanent and flexible contracts, and the distribution of vacancies over major occupation groups, by vacancies with wage information (column (1)) and without (column(2)). The third column calculates the difference between the two. Standard errors in parentheses.

Figure A.6: Distribution of vacancies and employment in Lightcast and ASHE data



Notes: Pooled data over 2014-2019. Occupations represent occupation groups at 1-digit SOC level.

Figure A.7: Representativeness of Lightcast Data over Time



Notes: Each data point is derived by regressing the number of vacancies (panel (a)) or median wage (panel (b)) in occupation-county cells in the Lightcast data on the corresponding number of employees in the ASHE data, within a given year, and collecting the R^2 . The flat profile of both figures suggests that the representativeness of Lightcast relative to ASHE stays constant over time.

D Natural Language Processing Details

This appendix provides a detailed description of the methodology used to construct our measures of flexible work arrangements based on annotations of vacancy text. We construct a ground-truth dataset by manually annotating job vacancy postings for schedule flexibility, remuneration type, and contract permanence. Annotated vacancies are drawn from random samples of Lightcast postings covering all years in the analysis period, with initial sampling stratified toward sectors in which flexible scheduling is relatively more prevalent, including education, retail, food services, and health care. Manual annotation was conducted by the authors in collaboration with a team of four trained research assistants, all of whom followed a common written annotation protocol. For schedule flexibility and contract permanence, the training dataset consists of 6,650 manually annotated vacancies. For remuneration type, which is comparatively easier to identify from vacancy text, the annotated sample comprises 1,192 postings. Although the full dataset is ultimately classified using a machine-learning model, the manually annotated vacancies play a central role in the analysis. They define the ground truth used to train the classifier, serve as the benchmark for model tuning, and enable the assessment of out-of-sample predictive accuracy. Manual annotation proceeded iteratively, with additional postings labelled until classifier performance stabilized across all dimensions.

All vacancy postings were coded using a detailed written annotation protocol that defined each classification category and specified the textual cues used to identify them. Research assistants classified vacancies at the posting level along four dimensions: (i) contract duration (permanent versus temporary), (ii) remuneration type (salaried versus non-salaried), (iii) hours information (zero-hours, part-time, full-time, or unspecified), and (iv) schedule flexibility (flexible, inflexible, or no schedule information). The protocol provided guidance on both explicit and implicit textual cues, including references to fixed shift times, rotating or variable schedules, on-call duties, zero-hours arrangements, and business-driven variability in hours or timing. It also included instructions for interpreting ambiguous formulations

commonly observed in vacancy text. Annotators applied the protocol consistently across postings, flagging uncertain cases for discussion and consensus coding. Given that annotation was conducted by multiple researchers, maintaining a shared understanding of the classification categories was essential; the written protocol played a central role in aligning interpretations and ensuring consistency across the annotation team.

We assessed the reliability of the manual annotations through a dual-coding and reconciliation exercise. A subset of 1,700 vacancy postings was independently coded by two researchers to evaluate inter-annotator consistency. Agreement rates for this sample were high, and reconciliation discussions, focused primarily on borderline or ambiguous cases, were used to refine and sharpen the operational definitions embedded in the final annotation protocol. The resulting ground-truth dataset is essential for accurate classification at scale. Simple keyword-based approaches, such as those employed in [duchini2020pay](#), are prone to both false positives (for example, uses of the term “flexible” that refer to worker traits rather than scheduling arrangements) and false negatives (such as flexible schedules described without canonical terminology). A supervised machine-learning approach trained on manually annotated data is therefore necessary to capture the contextual and semantic variation present in vacancy text.

D.1 LLM-based annotation procedure

To complement the manually annotated ground-truth dataset, we employ large language models (LLMs) to generate additional high-quality annotations for the schedule flexible category. These LLM-based labels serve both as a second perspective on the same underlying classification categories and as supplementary training data for the supervised classifiers applied to the full vacancy corpus. The manual, LLM-based, and model-generated classifications jointly anchor one another: each provides an independent yet related view of the conceptual constructs we seek to measure. In light of recent advances in LLM capabilities, LLM-based classifications offer a high-capacity and competitively accurate complement to

human annotation, providing an additional lens on the textual cues relevant to schedule flexibility. Their use allows us to construct a substantially larger and more representative annotated dataset than would be feasible through manual coding alone.³⁵

Because many indicators of schedule flexibility are expressed at the level of individual clauses or sentences, for example, short statements describing rotating shifts, variable hours, or on-call duties, we preprocess each vacancy posting by segmenting the text into sentences prior to classification. This sentence-level representation allows the classifier to evaluate relevant cues in isolation from unrelated contextual material within the same posting, reducing the risk that labels are driven by spurious co-occurrences elsewhere in the text. It also facilitates transparent validation of the resulting annotations, as correctness can be reviewed at the level of a single sentence rather than an entire posting.

Sentence segmentation is performed on the full corpus of over 60 million job vacancy postings using the *Stanza* natural language processing pipeline (version 1.10.1). *Stanza* tokenizes vacancy text into sentences prior to LLM-based annotation. Although computationally intensive, processing the UK sample alone required approximately 48 hours on 100 CPU cores, this approach yields substantial accuracy improvements relative to rule-based segmentation methods, as *Stanza*’s neural models incorporate contextual information when identifying sentence boundaries. No additional text preprocessing was applied prior to segmentation. Vacancy postings were supplied in their raw Lightcast format, as *Stanza* is designed to handle irregular punctuation and layout directly. Lightcast’s upstream preprocessing, including the removal of HTML artefacts and extraneous characters, further reduces noise in the input text. A manual spot check of 20 randomly selected vacancies indicates high segmentation accuracy. Observed deviations were minor and typically involved the occasional merging of multiple short sentences, most often arising from bullet-point formatting or unconventional punctuation.

³⁵Applying LLMs directly to the full set of vacancy postings would be prohibitively expensive at prevailing computational and usage costs in 2025, motivating their use as a targeted complement rather than a substitute for human classification.

D.1.1 Schedule-related sentences

The next stage of the pipeline constructs a sentence-level measure of schedule-related content. This classification acts as a pre-filter, as schedule flexibility is defined conditionally on the presence of schedule-related information at the sentence level. Implementing this step requires translating vacancy-level labels, assigned through manual annotation, into sentence-level classifications. To do so, we partition the 6,650 manually annotated job postings into two groups based on their vacancy-level schedule flexibility label: (i) postings classified as schedule flexible and (ii) postings classified as not schedule flexible. These two groups are processed differently for reasons of computational cost efficiency.

Because the number of postings classified as schedule flexible is relatively small, we apply API-based prompting using OpenAI’s *gpt-4o* model on this subset. We use *gpt-4o* at this stage because it offers substantially lower inference costs than reasoning-oriented models; candidate sentences identified in this step are subsequently re-evaluated using a more capable reasoning model. The role of the LLM at this stage is therefore limited to generating high-recall candidate labels, prioritizing coverage of potentially schedule-related sentences rather than final classification accuracy. Concretely, we take the 1,673 job postings manually classified as schedule flexible and apply a sentence-level prompt to each sentence within these postings to identify candidate instances of schedule-related content:

Does the following sentence from a job posting include any reference to the schedule of the job, even if it’s minor?

Schedule related information includes references to hours, schedules, shifts, weekend and bank holiday work, emergency work or overtime work, evening or night work, on-call duties, days, commission-based payments, job share, zero-hour or event-driven positions or "on-demand" style roles, time-windows. Answer with only 'yes' or 'no'. If unsure, answer 'yes'.

Sentence: {sentence}

Answer:

We use the LLM-generated sentence-level annotations as supervised training data to estimate a *fastText* classifier that predicts whether a sentence contains schedule-related information.³⁶ The classifier is trained on raw sentence text and produces a binary indicator for the presence of schedule-related content. Model hyperparameters are selected using five-fold cross-validation. We estimate a grid of *fastText* specifications that vary key training settings, including the learning rate, number of training iterations, embedding dimension, and the inclusion of word n-grams. The preferred specification is chosen to maximize average predictive accuracy across validation folds. This hyperparameter tuning procedure is applied consistently across all *fastText* models used in the paper, ensuring comparability across classification tasks. Unlike postings labeled as schedule flexible, which are processed using LLM prompting, postings labeled as not schedule flexible are instead screened using the *fastText* classifier. We apply the classifier to every sentence in postings labeled as not schedule flexible to detect schedule-related content within these vacancies, even though their vacancy-level label indicates no schedule flexibility. We pool the candidate schedule-related sentences identified through the LLM-based annotations and the *fastText* classifications. Finally, we apply a prompt at the sentence-level to the corresponding sentences:

Does this job posting contain information regarding the work schedule of the job role?

Relevant are:

Specific work hours;

Rotational shifts;

Night shifts;

Standard full-time schedules;

Shift times;

³⁶We use *fastText* `joulin2017bag`, a Python package developed by Meta, to perform fast supervised classification of textual data. *fastText* is extremely fast at inference, memory-efficient, trivial to retrain repeatedly as new annotations are added.

Flexible hours;

Fixed hours;

Overtime availability;

Emergency availability;

On-call duty.

These are relevant even if they are only occasionally required, such as a single mention of "may require overtime availability".

Not relevant are:

remote / hybrid work options;

length of the employment contract;

if it's a permanent or temporary contract.

Return as follows.

Row: [row number as stated]

Categorization: [yes/no]

Summary: [list all schedule related information verbatim].

Confirm that you understood the instructions with "ready" and await the job postings in one short sentence.

To generate the additional sentence-level annotations, we employ the highest-performing reasoning-oriented large language model available at the time of analysis (OpenAI's *o3-mini-high*). Sentences are provided to the model in batches of 200 through the ChatGPT interface, with the model processing sentences sequentially within each batch. To increase throughput, we run five batches in parallel, producing 1,000 annotated sentences per annotation round. Each batch requires up to approximately ten minutes of processing time, reflecting the computational cost of reasoning-based inference. Although inference is conducted through the ChatGPT interface under a Pro subscription, all prompts and model outputs are fully

logged. The resulting annotations are then treated as fixed training data and are used unchanged in all subsequent stages of the empirical pipeline.

We expand the sentence-level training data iteratively using an active-learning style procedure. In each round, we apply the trained *fastText* classifier to previously unseen vacancy text drawn from random samples spanning 2010–2020. From the resulting predictions, we retain only those sentences classified as schedule-related, thereby constructing a pool of high-probability candidate sentences. These candidate sentences are subsequently annotated using the *o3-mini-high* model to generate additional positively labeled training data. In total, we conduct four annotation rounds, each producing approximately 1,000–1,400 sentence-level annotations.

The final sentence-level training dataset consists of 8,399 sentences labeled as schedule-related and 43,649 sentences labeled as not schedule-related. We retain this class imbalance during training of the *fastText* classifier, rather than rebalancing the data, to acknowledge that the majority class exhibits important linguistic variety. The resulting classifier performs well, achieving a precision of 95%, a recall of 94%, and an F1 score of 95% on held-out validation data.

D.1.2 Schedule-flexible sentences

Having constructed a sentence-level classifier for schedule-related content, we next focus on schedule flexibility. We restrict attention to the set of sentences classified as schedule-related in the preceding stage of the pipeline and generate sentence-level schedule-flexibility labels using *o3-mini-high*. In this step, the model screens each schedule-related sentence for content specifically relevant to schedule flexibility. To classify schedule flexibility at the sentence level, we apply the following LLM-based annotation prompt to each sentence identified as schedule-related:

You are given a sequence of job posting texts. Determine if each references schedule flexibility [yes/no].

1. Core Definition

We define a schedule flexible work arrangement as one in which the timing of work is not fixed in the contract and is left open to future (possibly repeated) negotiations between the employer and the employee.

2. Nuances

Categorize a job as "schedule flexibility", if it mentions:

A work schedule that varies or rotates in a recurring or ongoing manner (rather than merely occasionally);

Different shifts (days/nights) on a regular rotation;

Shift allowance, that is additional pay for workers who work outside standard hours;

That shifts could vary, for example with "shifts TBC";

Weekend or Bank Holiday work requirements;

Alternative work shifts on offer, at least two, even if it does not state whether employees can negotiate their shifts;

The mere possibility of a shift change, without explicit mention of rotating shifts, variable schedules, negotiable hours, or regular shift changes, such as "to work x days out of 7 with one early and one late shift";

Occasional emergency or overtime calls (including potential overtime) qualify for schedule flexibility;

Shift type: Mixed, which suggests that hours can vary;

"Long hours available," or a reference of overtime expectations;

"Hours: To be confirmed," this suggests that working hours are not rigidly set and may be negotiated;

"Willing to work evenings if required," or mention of occasional overtime

Flexible-hours roles (explicitly stating "flexible hours," "adjustable schedule," or "choose your own hours");

"Flexibility is required..." although it does not say the schedule explicitly, but we can interpreted as schedule flexible broadly;

"a career that suits your working hours", which implies that the job is part of a career allows workers to align their work schedule with their personal preferences;

A role requiring continuous system availability 24/7/365, implying on-call duties and non-fixed working hours, qualifies as schedule flexible;

"..averaging x hours per week.", where the term "averaging" or "average" indicates that hours per week are changing, thus it is schedule flexible, the same holds if there are "on average x shifts";

implicit references to schedule flexibility, such as "flexible approach to support a 7-day week organisation";

"up to x hours per week", that is the "up to" suggest variations in the hours per week;

"x days per week" without specifying which days the work will be performed;

Commission-based payment structures with earning variability, such as "xxx per week average earnings"

A "job share", suggesting the role could be split between multiple people;

Includes both reactive flexibility (when it's necessary occasionally) and proactive flexibility (a structured, varied schedule);

Zero-hour or event-driven positions, or "on-demand" style roles;

Statements such as "able to work any day of the week" or that employees must be ready to work across multiple different time windows;

Employee benefits for flexible scheduling (e.g., "pick shifts around your s chedule," "roster your own hours," etc.);

A minor schedule flexible element, such as potential overtime, even if the job contains other suggestions of a fixed schedule in general;
Employer or employee control the working schedule & shifts.

3. Not Enough to Qualify

The following do not alone constitute schedule flexibility under these guidelines:

A role being temporary or part-time: Those alone are not schedule flexibility.

Merely stating the occupation (e.g., hospitality, chef) does not automatically qualify as flexible or rotating unless the post explicitly references a rotating or flexible schedule.

Traveling or lodging away

Return as follows.

Row: [row number]

Categorization: [categorization yes/no]

Explanation: [short explanation]

Return no additional text before and after.

Confirm that you understood the instructions and await the job postings in one short sentence.

We further expand the sentence-level training dataset through additional annotation rounds designed to improve the performance of the schedule-flexibility classifier. In each round, we randomly sample job postings from the period 2012–2020 and apply the *fast-Text* model, trained on the currently available annotations, to classify sentences for schedule flexibility. From this output, we retain only sentences predicted to be relevant to schedule

flexibility, thereby concentrating annotation effort on likely positive examples. Each additional annotation round yields approximately 1,000–2,000 newly labeled sentences. In total, the final training dataset comprises 49,907 annotated sentences, of which 6,703 are labeled as containing references to schedule flexibility.

D.2 Natural Language Processing Details

The use of LASSO regularization is associated with a tuning parameter that controls the strength of regularization. It’s choice can change the accuracy of the classification model, by trading-off bias and variance. It is used within the log-likelihood of the logistic regression model:

$$\sum_{i=1}^D \log \Pr(y_{i,k}|x_i) - \frac{\lambda}{2} \|\beta_k\|_1. \quad (\text{D.1})$$

Algorithm 1 describes the procedure to choose the tuning parameter λ with the best out-of-sample prediction performance. We use the balanced F-measure to evaluate the accuracy of our classification algorithm. This equally weights precision and recall using the harmonic mean. The precision score penalises false positives (irrelevant items retrieved) and the recall score penalises false negatives (relevant items that are not retrieved).

$$\text{Balanced F-measure} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (\text{D.2})$$

where

$$\text{Precision} = \frac{\sum_i (\hat{y}_i = 1 \ \& \ y_i = 1)}{\sum_i (\hat{y}_i = 1 \ \& \ y_i = 1) + \sum_i (\hat{y}_i = 1 \ \& \ y_i = 0)} \quad (\text{D.3})$$

$$\text{Recall} = \frac{\sum_i (\hat{y}_i = 1 \ \& \ y_i = 1)}{\sum_i (\hat{y}_i = 1 \ \& \ y_i = 1) + \sum_i (\hat{y}_i = 0 \ \& \ y_i = 1)} \quad (\text{D.4})$$

with $\hat{y}_i$ giving the predicted label of vacancy i and y_i giving the true label.

In summary, we determine the tuning parameter by a grid search over a 5-fold cross-

validated F -score. Consider a work arrangement c . For a given value of the tuning parameter λ , we randomly sample a balanced set of vacancies. from the training set, $\mathbb{D}_S$. We split $\mathbb{D}_S$ into five equally sized groups $\mathcal{S}_j$ for $j = 1, \dots, 5$. For each group $\mathcal{S}_\ell$, we select the parameter, $\hat{\beta}_{k,\lambda,\ell}$ that maximises the objective function at Equation D.1 on the subset of vacancies excluding group $\mathcal{S}_\ell$, i.e. $\mathbb{D}_S \setminus \mathcal{S}_\ell$. $\hat{\beta}_{k,\lambda,\ell}$ is then used to predict the presence of work vacancy c in the excluded group $\mathcal{S}_\ell$. The model’s predictions are compared to the true vacancy labels for vacancies in $\mathcal{S}_\ell$ and the accuracy of the model evaluated (see below). We repeat this procedure 500 times for each value of λ for each work arrangement. We select the value of λ that maximises the average of the cross-validated F -score over the 500 sampling draws.

Algorithm 2 describes how the trained logistic-regression classifier is applied to a new vacancy document d . For each vacancy d , we construct a document vector x_i using the vocabulary V and the same feature-engineering procedure as in the training stage. The classifier then evaluates each work-arrangement class $k \in \mathbb{C}$ separately. Using the class-specific coefficient vector $\hat{\beta}_k^*$ contained in the weight tuple W obtained from Algorithm 1, we compute the predicted probability that document d belongs to class k . A class is assigned whenever this probability exceeds the decision threshold of 0.5, yielding a (potentially multi-label) assignment of work-arrangement categories to each vacancy based on its textual content.

Algorithm 1: Training the Logistic Regression model $(\mathbb{C}, \mathbb{D})$

Result: Return the tuple $W = (\hat{\beta}_1^*, \hat{\beta}_2^*, \dots, \hat{\beta}_{5000}^*)$ containing the vocabulary weights

for each class $k \in \mathbb{C}$

extract the vocabulary in $\mathbb{D}$

for $k \in \mathbb{C}$ **do**

for $\lambda \in [0.01, 0.05]$ **do**

for $r = 1, \dots, 500$ **do**

 compute as N_k the number of vacancies in class k which are labelled with 1

 sample N_k vacancy texts labelled with 0 and the same number of text labelled with 1 and call the sample $\mathbb{D}_S$

 randomly divide documents in $\mathbb{D}_S$ into 5 equally sized blocks $\mathcal{S}_\ell$, for $\ell = 1, \dots, 5$

for $\ell = 1, \dots, 5$, **do**

 find $\hat{\beta}_{k,\lambda,\ell}$ which minimises

$$- \sum_{i \in \mathbb{D}_S \setminus \mathcal{S}_\ell} \log \Pr(y_{i,k} | x_i) + \lambda \|\beta_{k,\lambda,\ell}\|_1$$

 where $\|\cdot\|_1$ is the 1-norm while excluding the observations in block ℓ

 compute the cross-validated balanced F -measure, as defined in Equation (D.2), using observations in block ℓ and $\hat{\beta}_{k,\lambda,\ell}$

end

end

end

 select the λ with the highest average cross-validated balanced F -measure and call the corresponding estimate $\hat{\beta}_k^*$

end

Algorithm 2: Apply the Logistic Regression model $(\mathbb{C}, V, W, d)$

Result: Assign class $k \in \mathbb{C}$ to vacancy i

construct a vector representation x_i according to the vocabulary

for $k \in \mathbb{C}$ **do**

 compute the probability $\Pr(y_{i,k} = 0|x_i)$ and $\Pr(y_{i,k} = 1|x_i)$ using the coefficients

$\hat{\beta}_k^*$ in weight vector W as

$$\Pr(y_{i,k} = 1|x_i) = \frac{\exp(x_i' \hat{\beta}_k^*)}{1 + \exp(x_i' \hat{\beta}_k^*)}, \quad \Pr(y_{i,k} = 0|x_i) = 1 - \Pr(y_{i,k} = 1|x_i)$$

if $\Pr(y_{i,k} = 1|x_i) > 0.5$ **then**

 | assign class k to vacancy i

end

end
